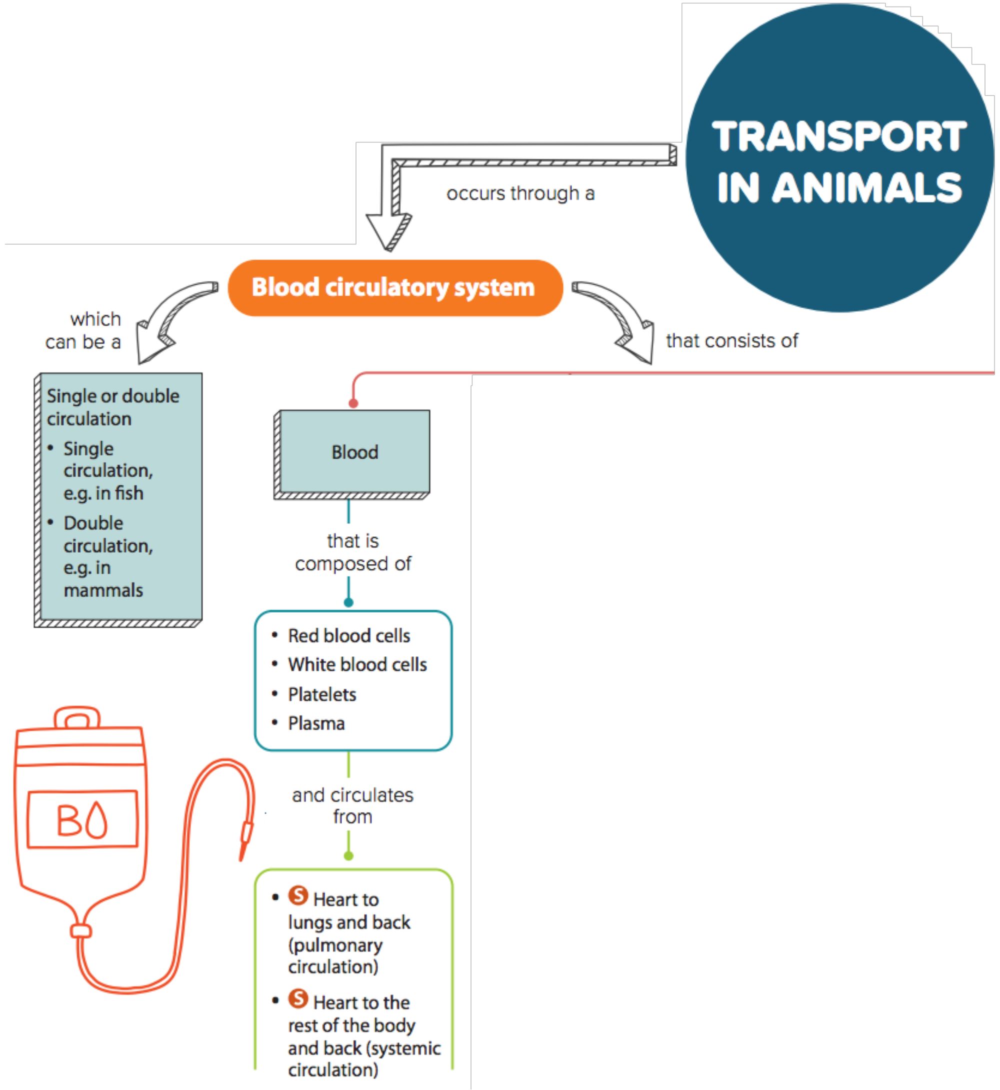
CHAPTER 9

TRANSPORT IN ANIMALS

9.1 Transport System: Structure and Composition of Blood

In this section, you will learn the following:

- Describe the circulatory system as a system of blood vessels with a pump and valves to ensure one-way flow of blood.
- List the components of blood.
- Identify red and white blood cells in photomicrographs and diagrams.
- State the functions of red blood cells, white blood cells, platelets and plasma.
- S Identify lymphocytes and phagocytes in photomicrographs and diagrams.
- S State the functions of lymphocytes and phagocytes.
- State the roles of blood clotting.
- S Describe the process of clotting.



9.1 Circulatory systems

Core

1 Describe the circulatory system as a system of blood vessels with a pump and valves to ensure one-way flow of blood

Supplement

2 Describe the single circulation of a fish

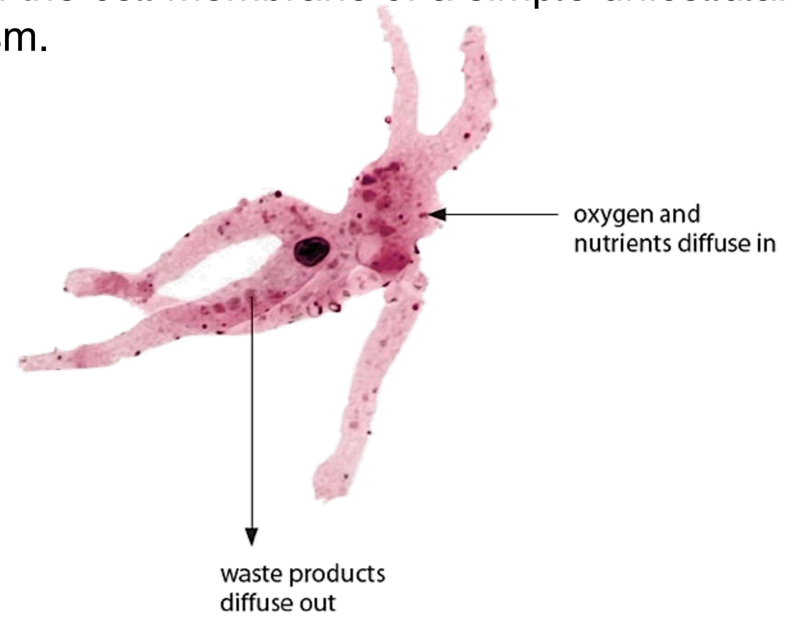
3 Describe the double circulation of a mammal

4 Explain the advantages of a double circulation

Transportation of substance in animals

Simple unicellular organism

Oxygen, nutrients and waste products diffuse easily through the cell membrane of a simple unicellular organism.



A highly magnified unicellular organism

LINK

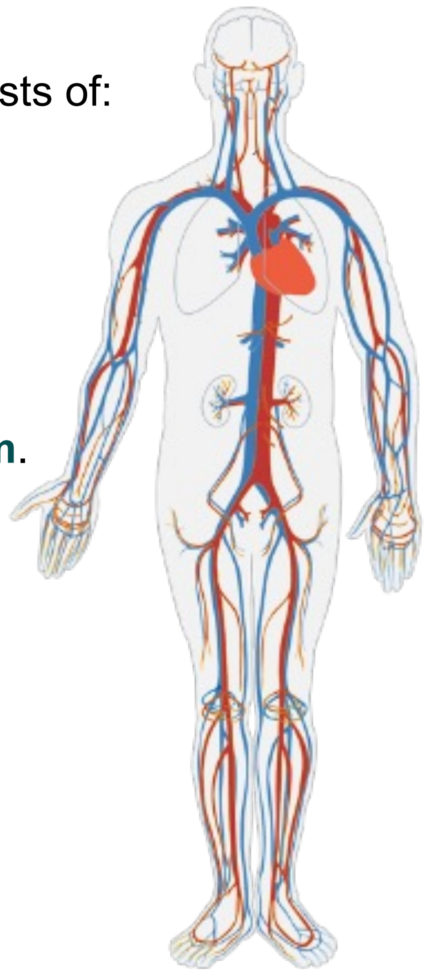
What is diffusion? Recall what you have learnt in Chapter 3.

Complex multicellular organism

Transport system in mammals consists of:

- Blood vessels
- Blood
- Heart
- Valves

The main transport system in humans is the **circulatory system**.



The human circulatory system

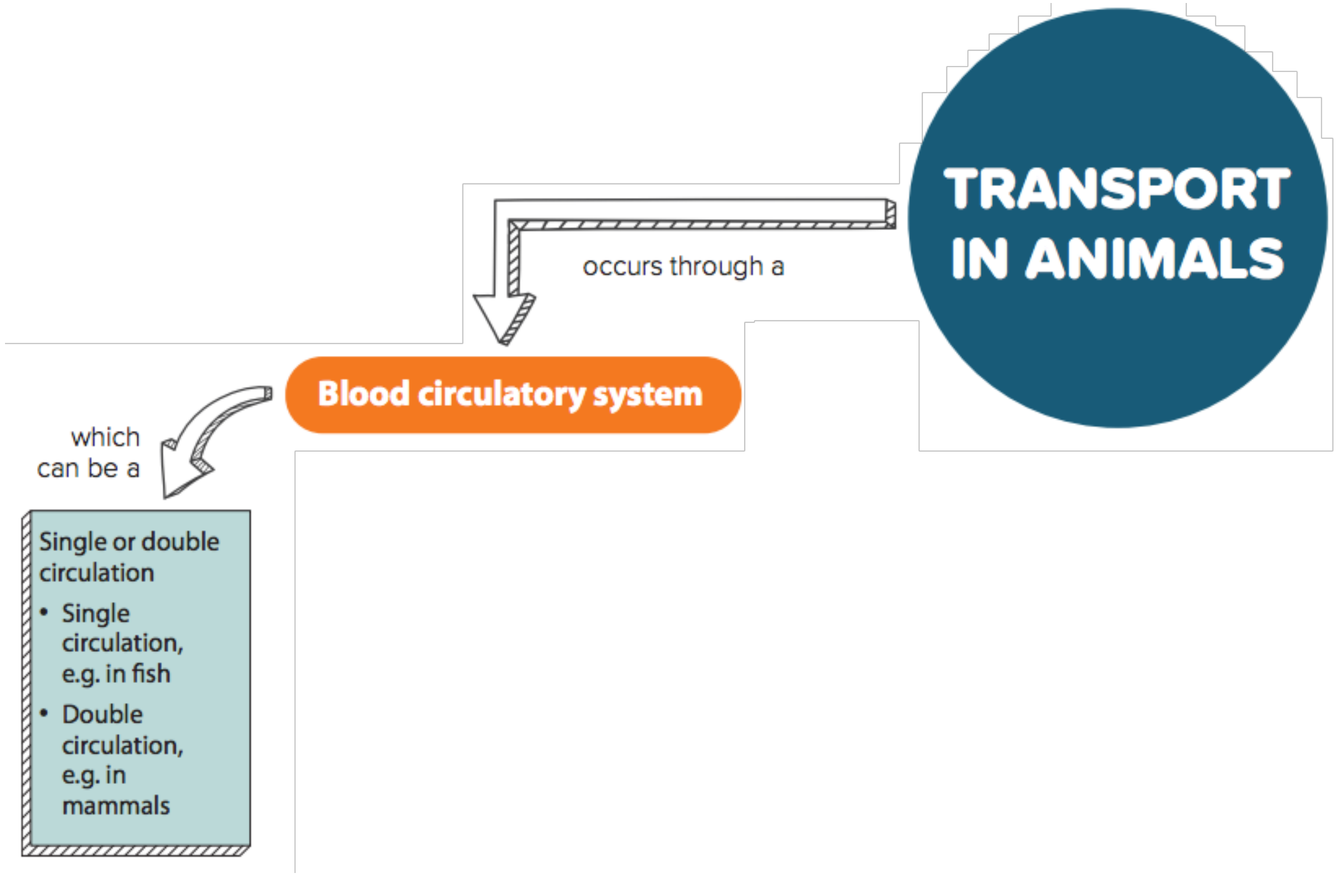
Let's Investigate 9A

- Drawing of a red blood cell should look like the red blood cell in Figures 9.2 and 9.4.
- Drawings of different types of white blood cells should look like the white blood cells in Figures 9.2 and 9.4.

9.3 Transport in Animals

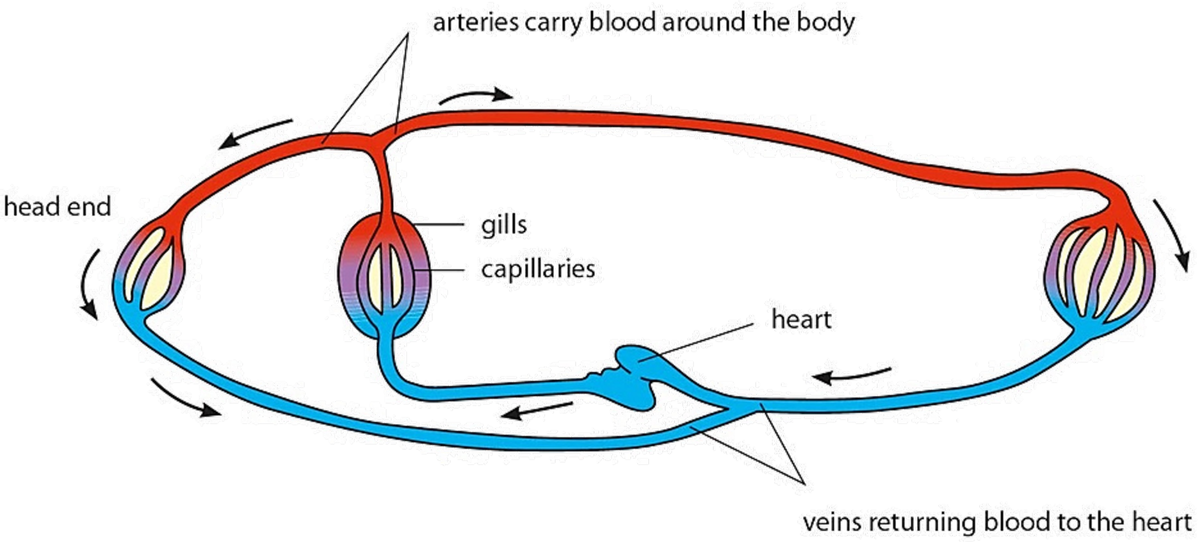
In this section, you will learn the following:

- Describe the single circulation of fish.
- Describe the double circulation of mammals.
- Explain the advantages of a double circulation.



Single circulation of fish

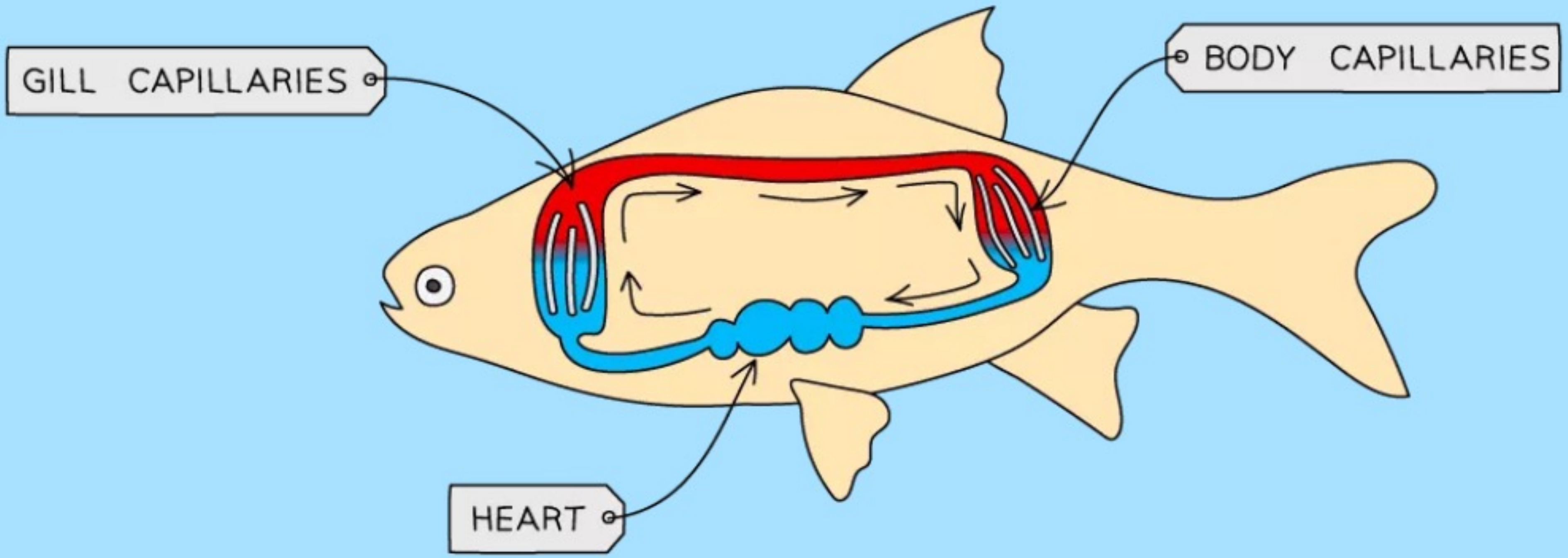
In some animals such as fish, blood flows through the heart once during each circuit of the body → **single circulation**.

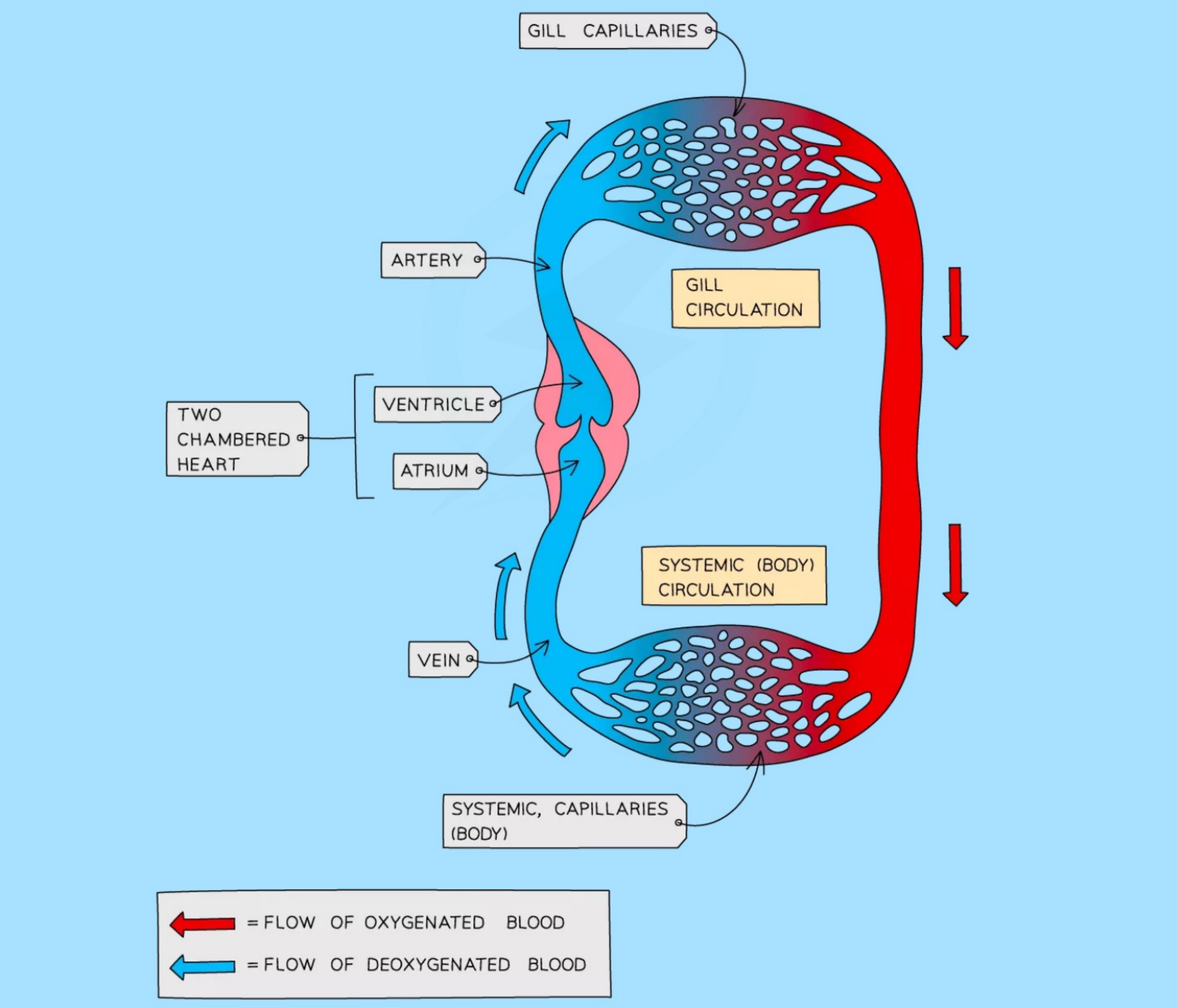


Single circulation in a fish

Circulation in Different Animals

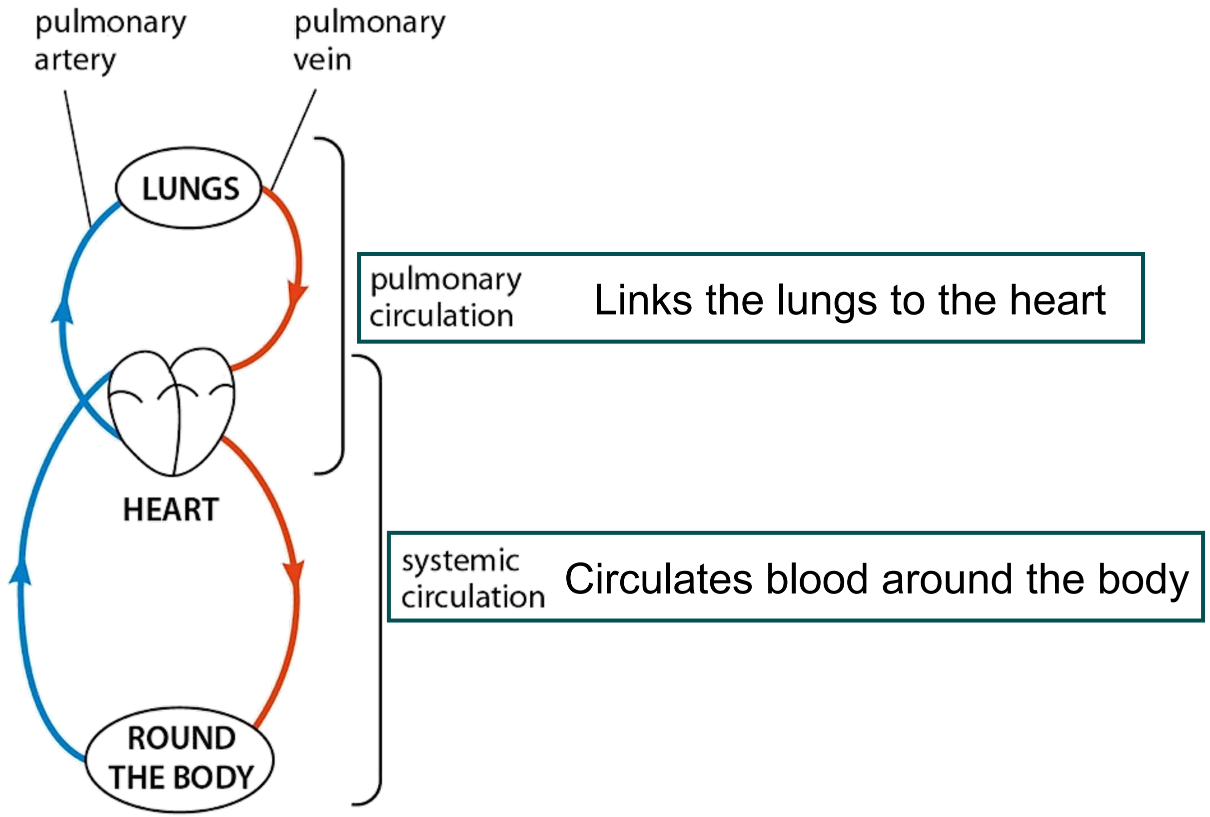
- Fish have a **two-chambered heart** and a **single circulation**
- This means that **for every one circuit of the body, the blood passes through the heart once**




GILL CAPILLARIES
ARTERY
GILL CIRCULATION
TWO CHAMBERED HEART
VENTRICLE
ATRIUM
SYSTEMIC (BODY) CIRCULATION
VEIN
SYSTEMIC, CAPILLARIES (BODY)
= FLOW OF OXYGENATED BLOOD
= FLOW OF DEOXYGENATED BLOOD

Double circulation of mammals

In mammals, blood flows through the heart twice during each circuit of the body → **double circulation**.
Double circulation in mammals consists of the **pulmonary circulation** & the **systemic circulation**.

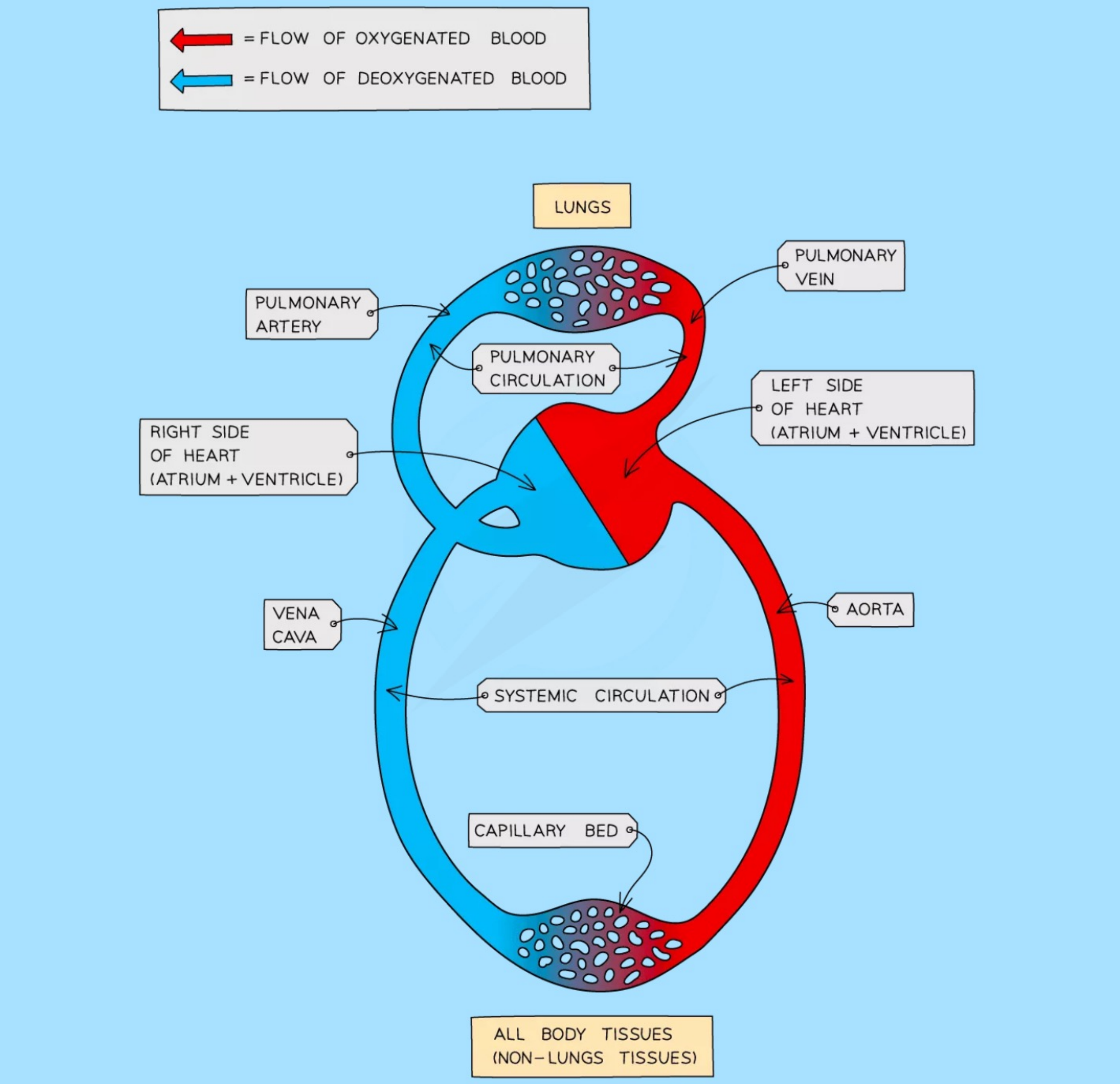


Double circulation in a mammal

LINK

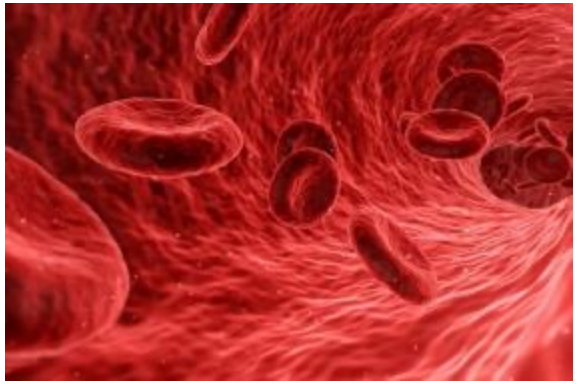
Why do mammals need to maintain a relatively high metabolic rate compared to organisms like fish? Find out in Chapter 15.

- Mammals have a **four-chambered heart** and a **double circulation**
- This means that **for every one circuit of the body, the blood passes through the heart twice**
- The right side of the heart receives **deoxygenated blood** from the body and **pumps it to the lungs** (the **pulmonary circulation**)
- The left side of the heart receives **oxygenated blood** from the lungs and **pumps it to the body** (the **systemic circulation**)

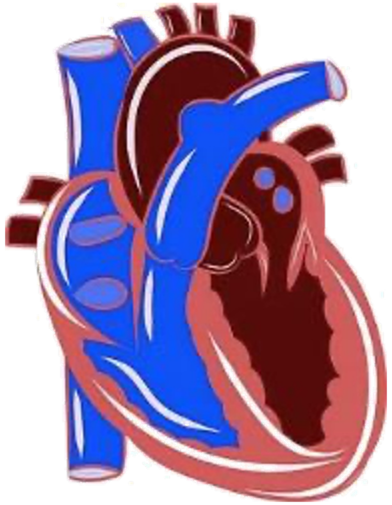

= FLOW OF OXYGENATED BLOOD
= FLOW OF DEOXYGENATED BLOOD
LUNGS
PULMONARY ARTERY
PULMONARY VEIN
PULMONARY CIRCULATION
LEFT SIDE OF HEART (ATRIUM + VENTRICLE)
RIGHT SIDE OF HEART (ATRIUM + VENTRICLE)
VENA CAVA
AORTA
SYSTEMIC CIRCULATION
CAPILLARY BED
ALL BODY TISSUES (NON-LUNGS TISSUES)

Advantages of a double circulation

- Blood flows more slowly through the lungs, allowing sufficient time for it to be well oxygenated.



- Four-chambered heart ensures separation of oxygenated blood from deoxygenated blood.



- Oxygenated blood is distributed to the body tissues more quickly.

Advantages of a Double Circulation

- Blood travelling through the small capillaries in the lungs **loses a lot of pressure** that was given to it by the pumping of the heart, meaning it **cannot travel as fast**
- By returning the blood to the heart after going through the lungs its **pressure can be raised again** before sending it to the body, meaning **cells** can be supplied with the **oxygen and glucose** they need for respiration **faster and more frequently**

9.4 Blood

Core

1 List the components of blood as: red blood cells, white blood cells, platelets and plasma

2 Identify red and white blood cells in photomicrographs and diagrams

3 State the functions of the following components of blood:

(a) red blood cells in transporting oxygen, including the role of haemoglobin

(b) white blood cells in phagocytosis and antibody production

(c) platelets in clotting (details are **not** required)

(d) plasma in the transport of blood cells, ions, nutrients, urea, hormones and carbon dioxide

Supplement

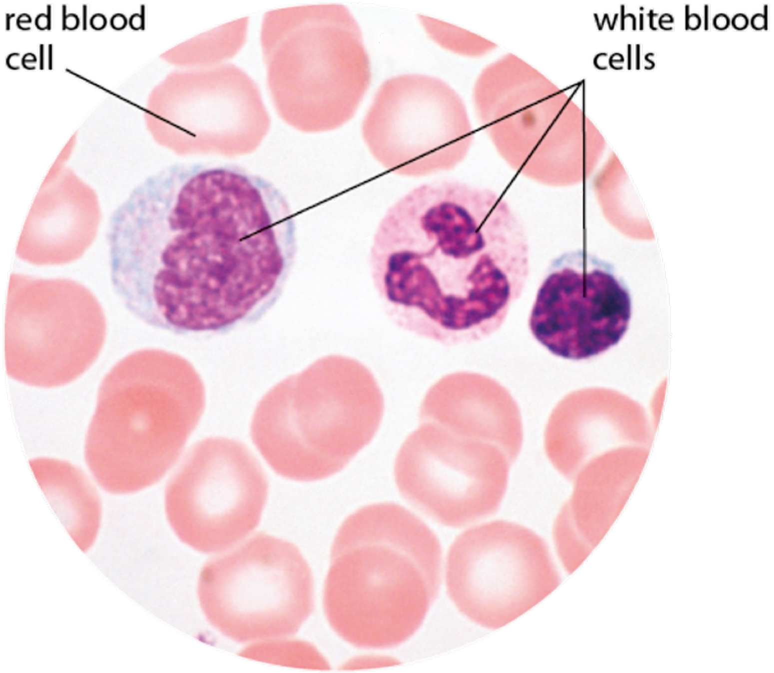
5 Identify lymphocytes and phagocytes in photomicrographs and diagrams

6 State the functions of:

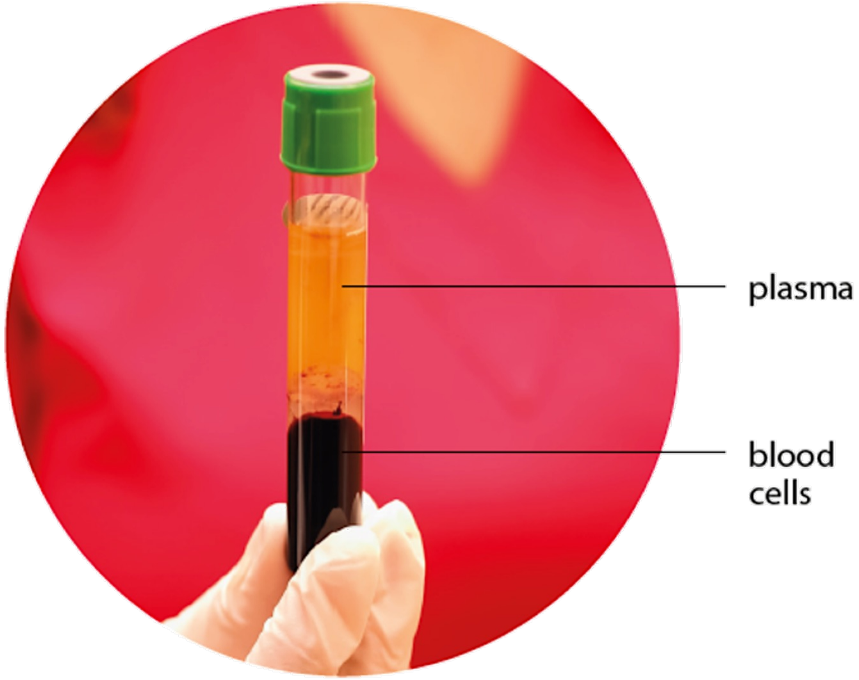
(a) lymphocytes – antibody production

(b) phagocytes – engulfing pathogens by phagocytosis

Blood as a fluid tissue



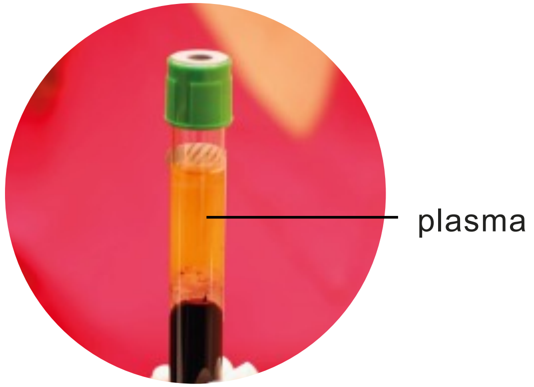
Blood smear under a light microscope



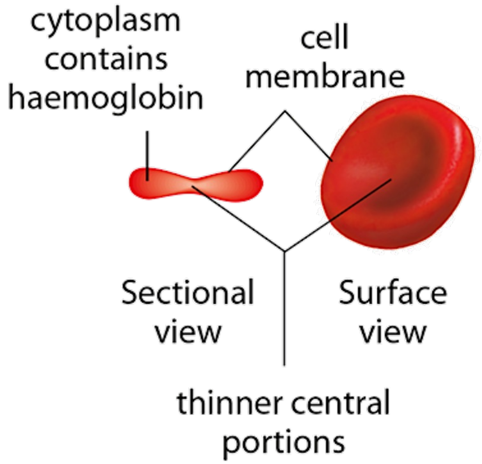
Blood spun at high-speed can be separated out into layers

Main components of blood

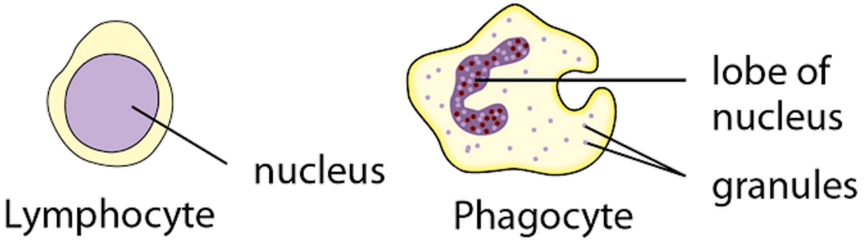
1. Plasma



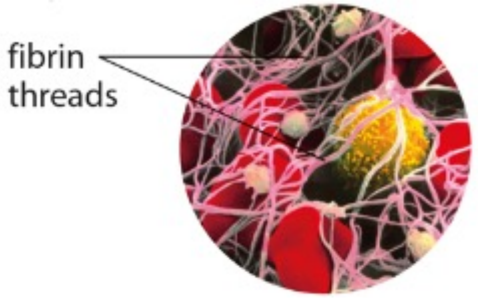
2. Red blood cells



3. White blood cells

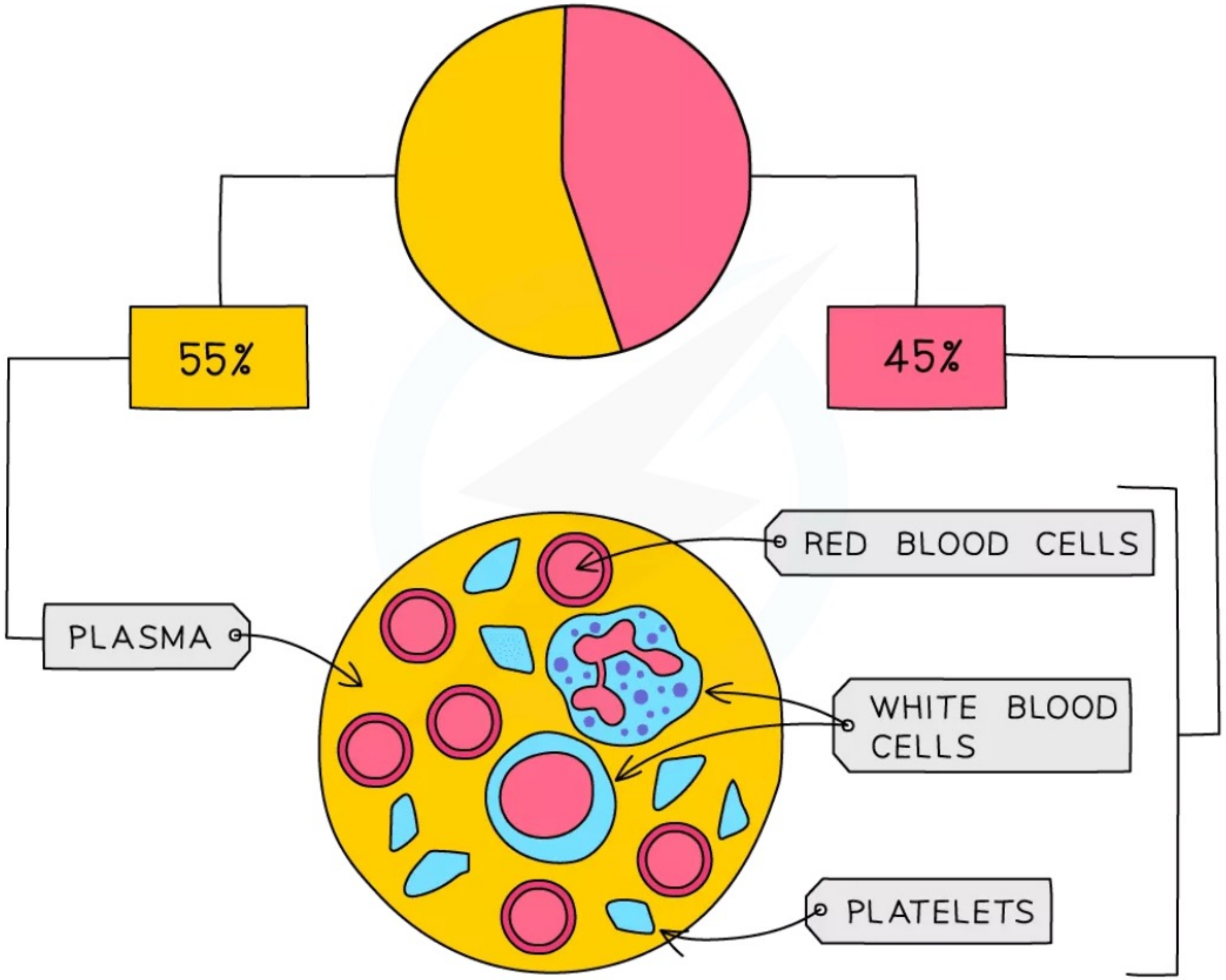


4. Platelets



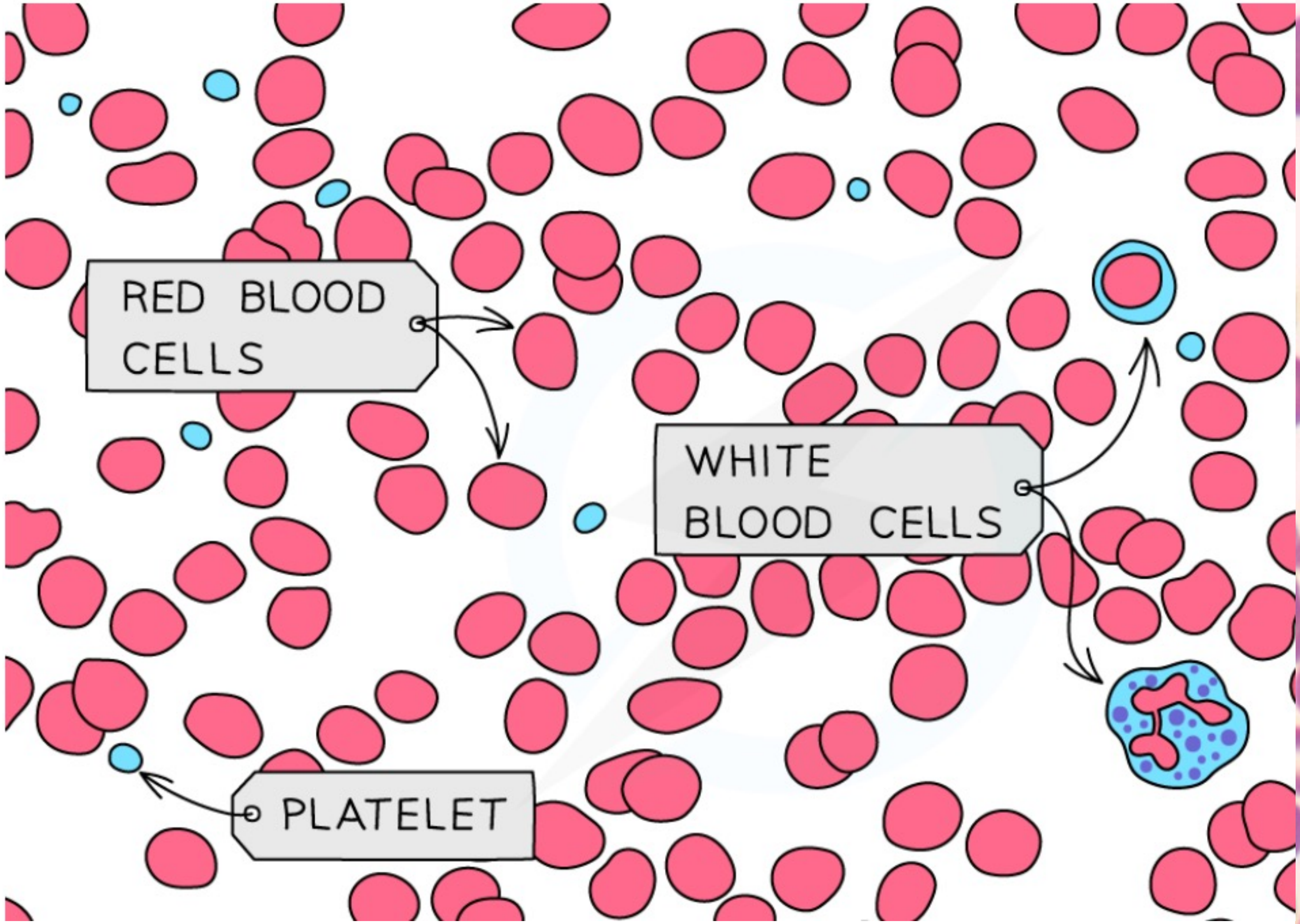
Components of Blood

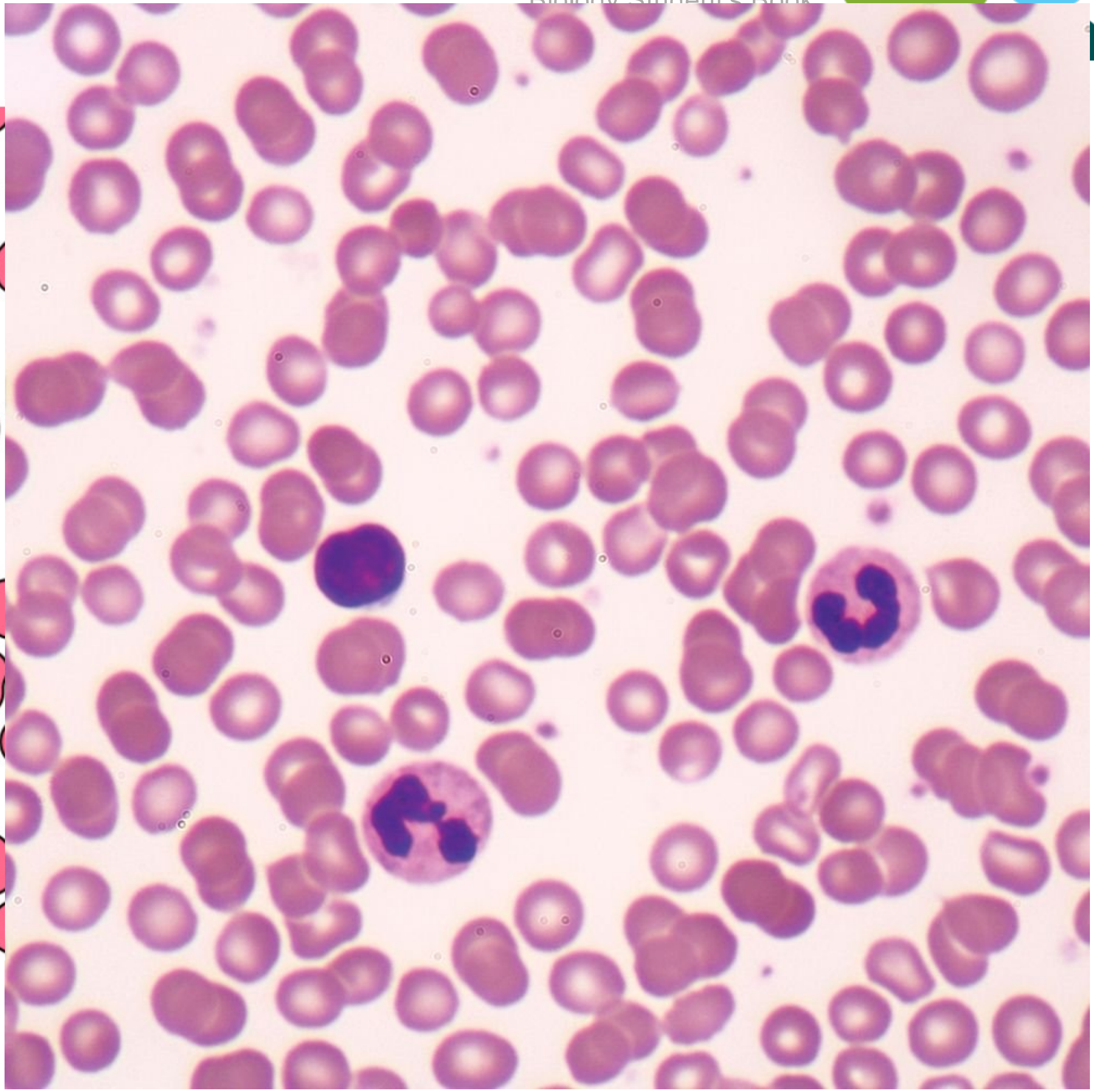
- Blood consists of **red blood cells, white blood cells, platelets and plasma**



Components of the blood:

COMPONENT	STRUCTURE
RED BLOOD CELLS	BICONCAVE DISCS CONTAINING NO NUCLEUS BUT PLENTY OF THE PROTEIN HAEMOGLOBIN
WHITE BLOOD CELLS	LARGE CELLS CONTAINING A BIG NUCLEUS, DIFFERENT TYPES HAVE SLIGHTLY DIFFERENT STRUCTURES AND FUNCTIONS
PLATELETS	FRAGMENTS OF CELLS
PLASMA	STRAW COLOURED LIQUID


RED BLOOD CELLS
WHITE BLOOD CELLS
PLATELET

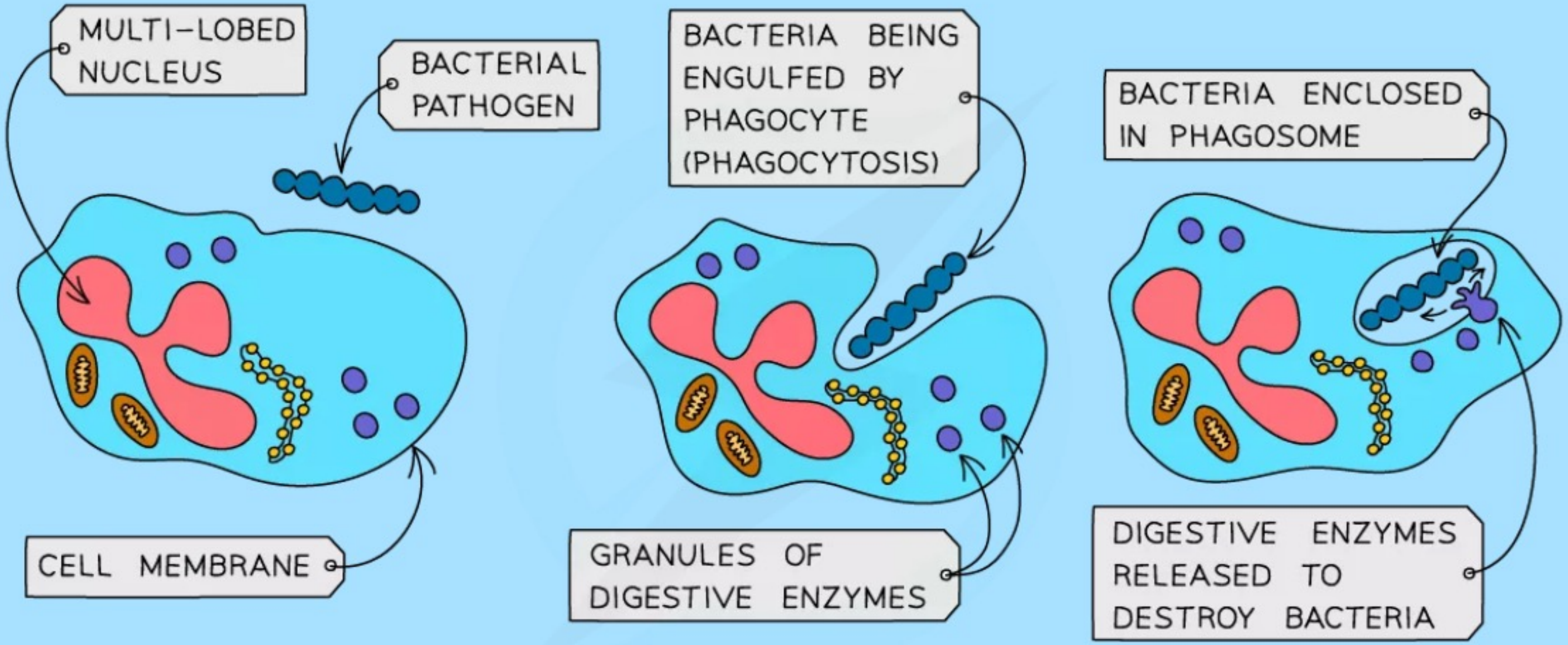


Types of White Blood Cell

- White blood cells are part of the body's **immune system**, defending against infection by pathogenic microorganisms
- There are two main types, **phagocytes and lymphocytes**

Phagocytes

- Carry out **phagocytosis** by **engulfing and digesting pathogens**



- Phagocytes have a sensitive cell surface membrane that can detect chemicals produced by pathogenic cells
- Once they encounter the pathogenic cell, they will engulf it and **release digestive enzymes** to digest it
- They can be easily recognised under the microscope by their **multi-lobed nucleus** and their **granular cytoplasm**

Lymphocytes

- Produce **antibodies** to destroy pathogenic cells and **antitoxins** to neutralise toxins released by pathogens
- They can easily be recognised under the microscope by their **large round nucleus** which takes up nearly the whole cell and their **clear, non-granular cytoplasm**

Functions of the Parts of the Blood

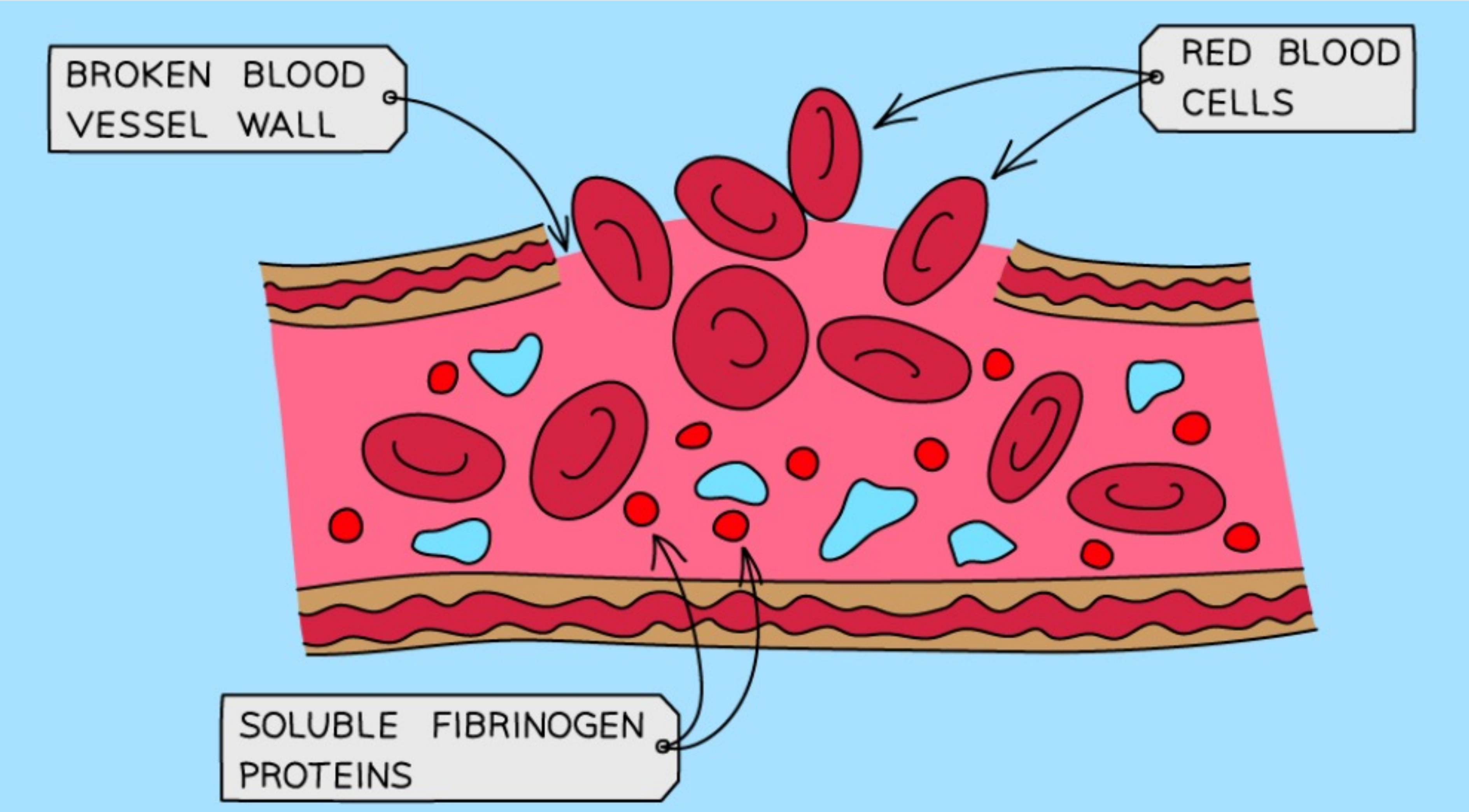
- **Plasma** is important for the transport of **carbon dioxide, digested food (nutrients), urea, mineral ions, hormones and heat energy**
- **Red blood cells transport oxygen around the body** from the lungs to cells which require it for aerobic respiration
- They carry the oxygen in the form of **oxyhaemoglobin**
- **White blood cells** defend the body against infection by pathogens by carrying out **phagocytosis and antibody production**
- **Platelets** are involved in **helping the blood to clot**

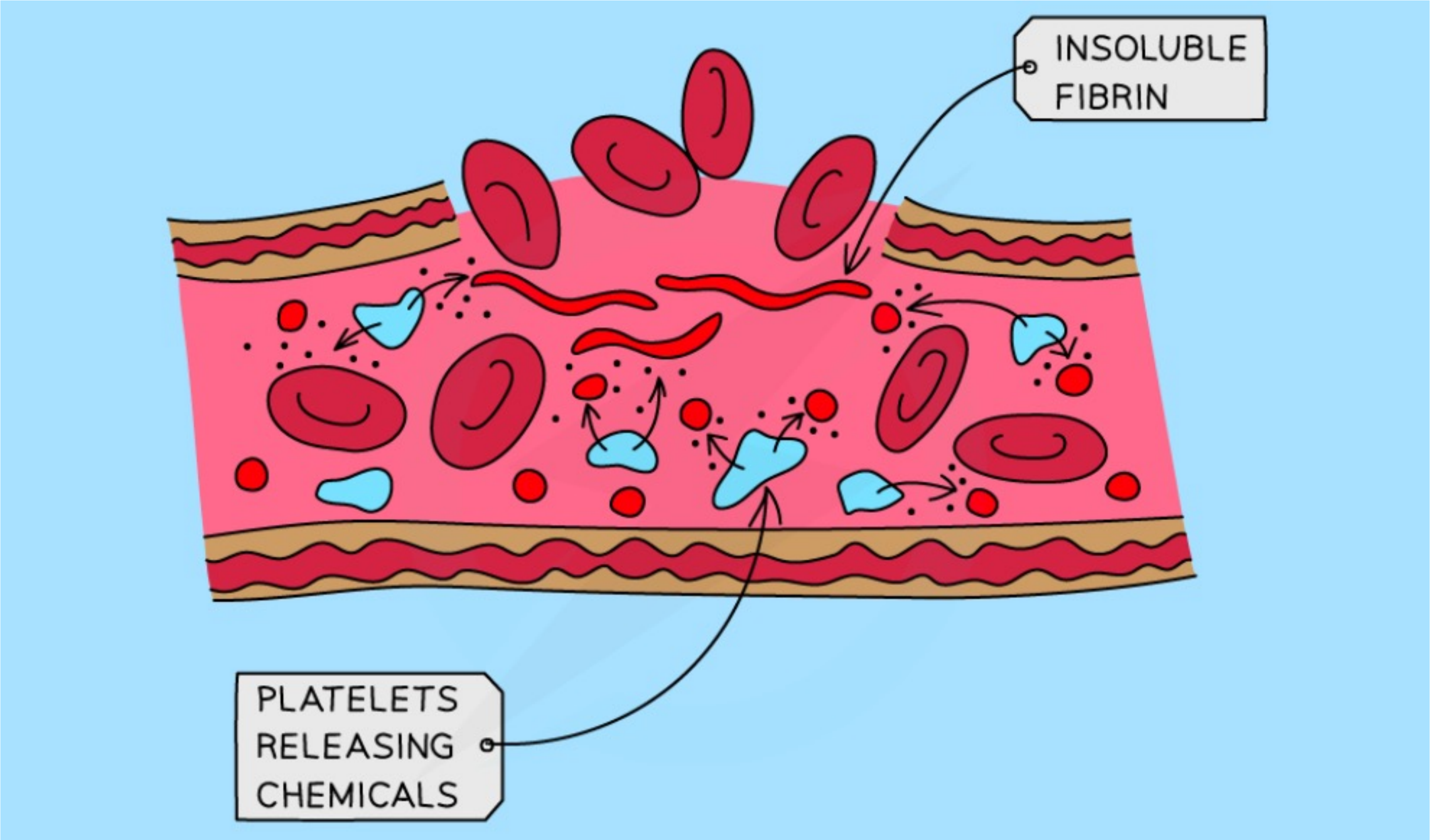
4 State the roles of blood clotting as preventing blood loss and the entry of pathogens

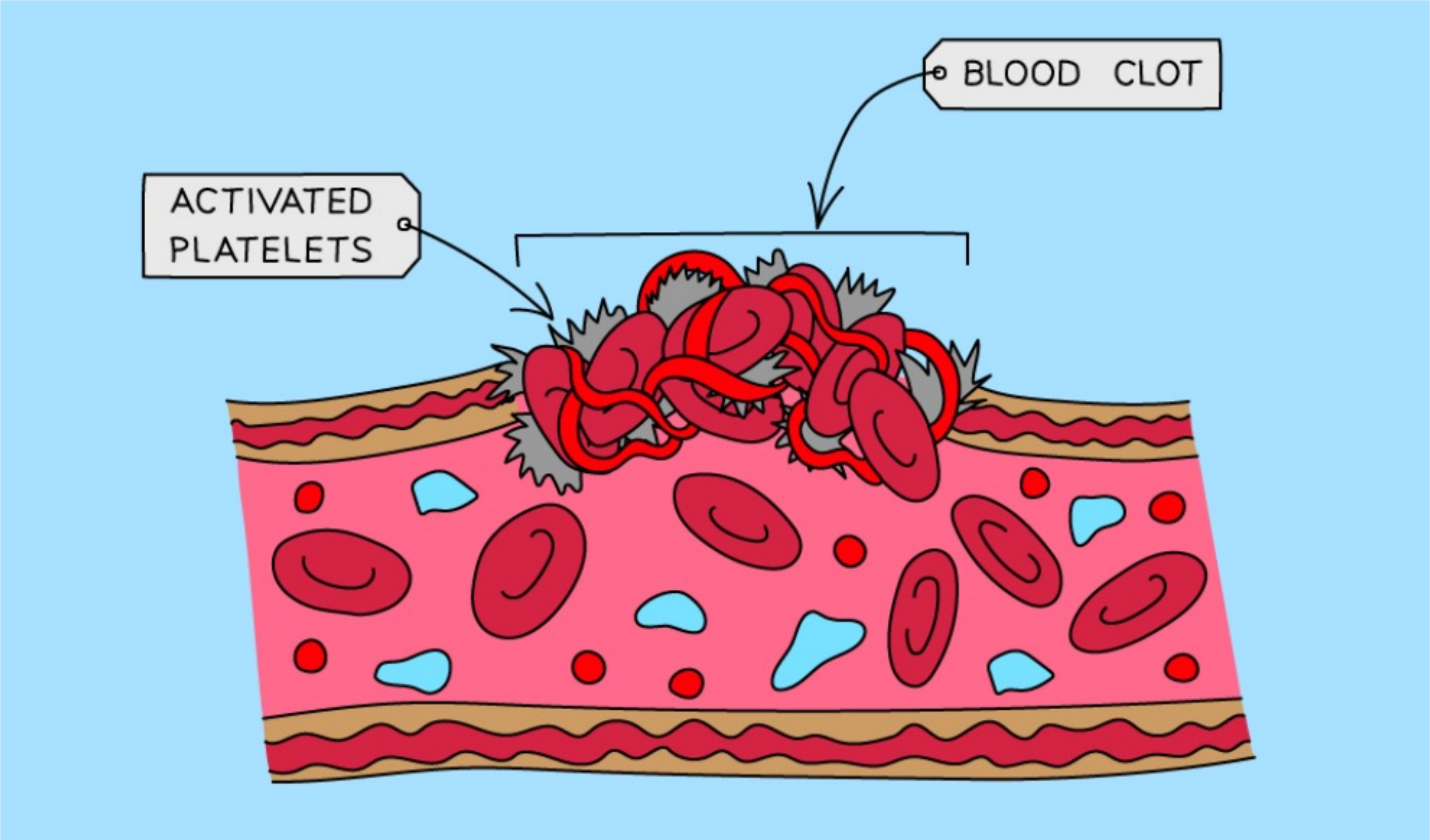
7 Describe the process of clotting as the conversion of fibrinogen to fibrin to form a mesh

Blood Clotting

- Platelets are **fragments of cells which are involved in blood clotting** and forming scabs where skin has been cut or punctured
- Blood clotting **prevents continued / significant blood loss** from wounds
- Scab formation seals the wound with an insoluble patch that **prevents entry of microorganisms** that could cause infection
- It remains in place until new skin has grown underneath it, sealing the skin again


BROKEN BLOOD VESSEL WALL
RED BLOOD CELLS
SOLUBLE FIBRINOGEN PROTEINS

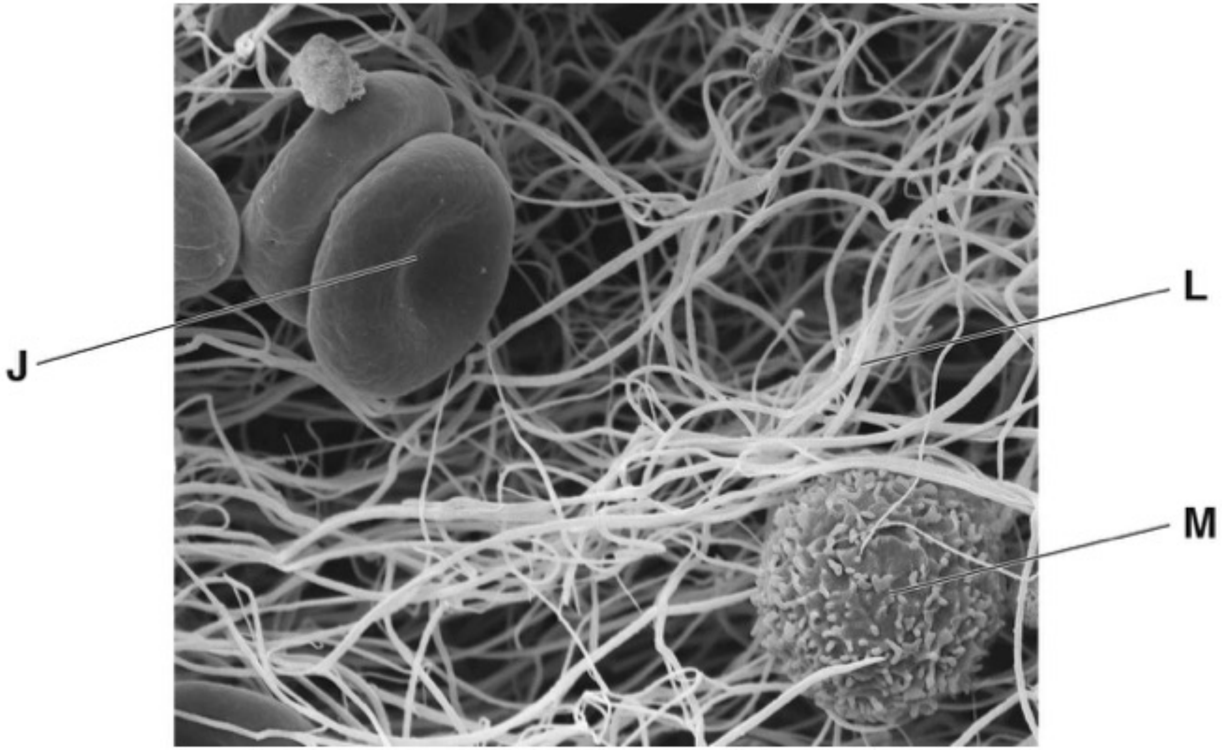

INSOLUBLE FIBRIN
PLATELETS RELEASING CHEMICALS


BLOOD CLOT
ACTIVATED PLATELETS

How the blood clots

- When the skin is broken (i.e. there is a wound) platelets arrive to stop the bleeding
- A series of reactions occur within the blood plasma
- Platelets release chemicals that cause **soluble fibrinogen proteins** to convert into **insoluble fibrin** and form an **insoluble mesh** across the wound, trapping red blood cells and therefore **forming a clot**
- The clot eventually dries and develops into a **scab** to protect the wound from bacteria entering

The photomicrograph is of a blood clot.



Describe how a blood clot forms.

Use the letters in the photomicrograph in your answer.

any five from:
ref. to platelets ;

fibrinogen is converted to fibrin / **L** ;

fibrinogen is soluble / fibrin is insoluble ;

(**L** / fibrin) forms a, mesh / AW ;

(**L** / fibrin) traps / AW, blood cells / **J** / **M** ;

J is a red (blood) cell ;

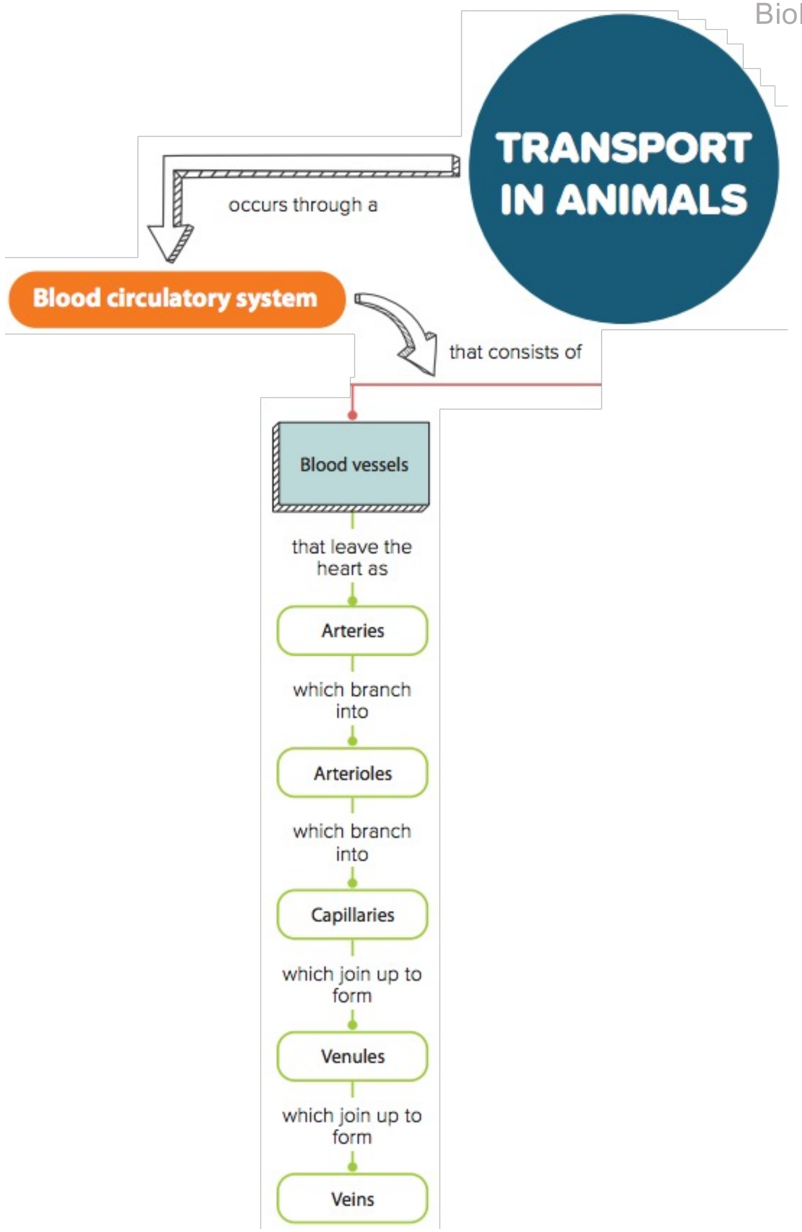
L is fibrin ;

M is a, white (blood) cell / lymphocyte / phagocyte ;

9.2 The Circulatory System

In this section, you will learn the following:

- State the functions of capillaries.
- S Explain how the structure and function of capillaries are related.
- State that blood from the heart is pumped away through arteries and returns to the heart through veins.
- Describe the structure of arteries, veins and capillaries.
- S Explain how the structure of arteries and veins relates to the pressure of the blood that they transport.



9.3 Blood vessels

Core

1 Describe the structure of arteries, veins and capillaries, limited to: relative thickness of wall, diameter of the lumen and the presence of valves in veins

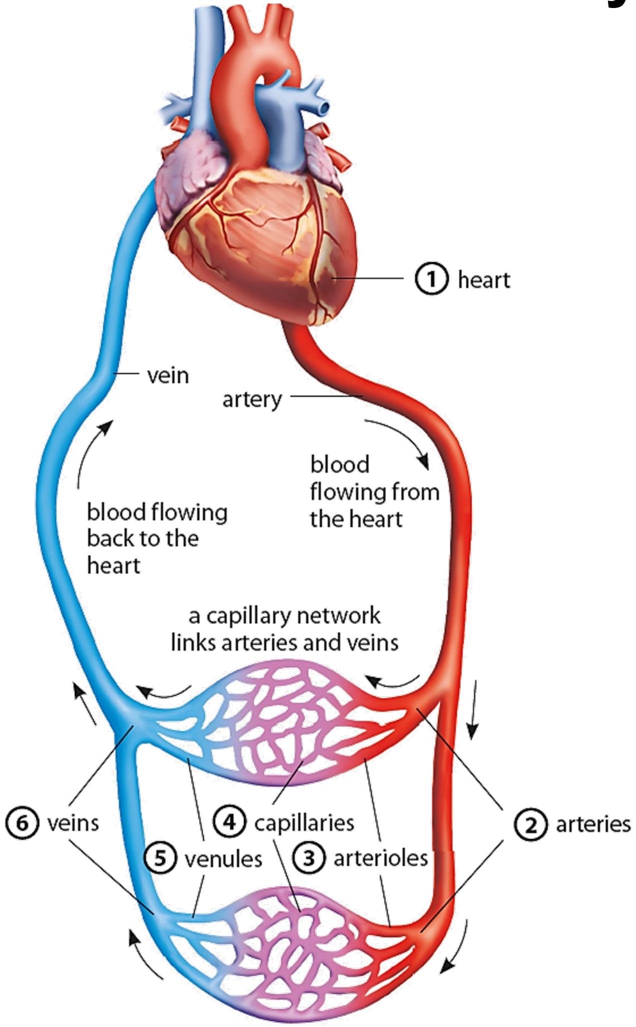
2 State the functions of capillaries

Supplement

4 Explain how the structure of arteries and veins is related to the pressure of the blood that they transport

5 Explain how the structure of capillaries is related to their functions

Components of the Circulatory System

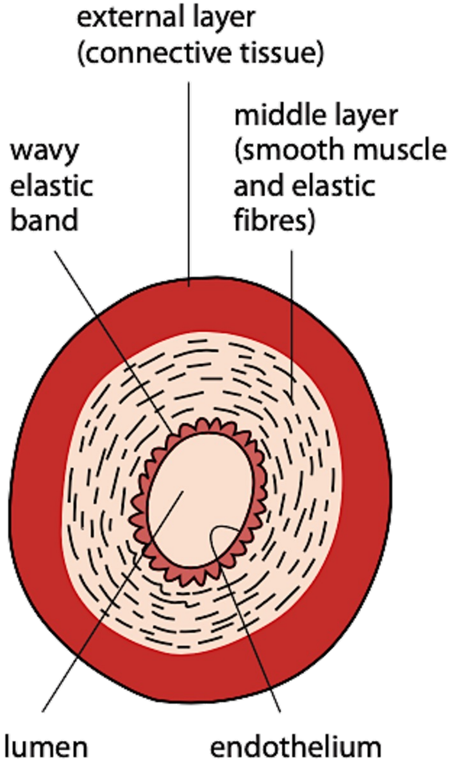


The circulatory system

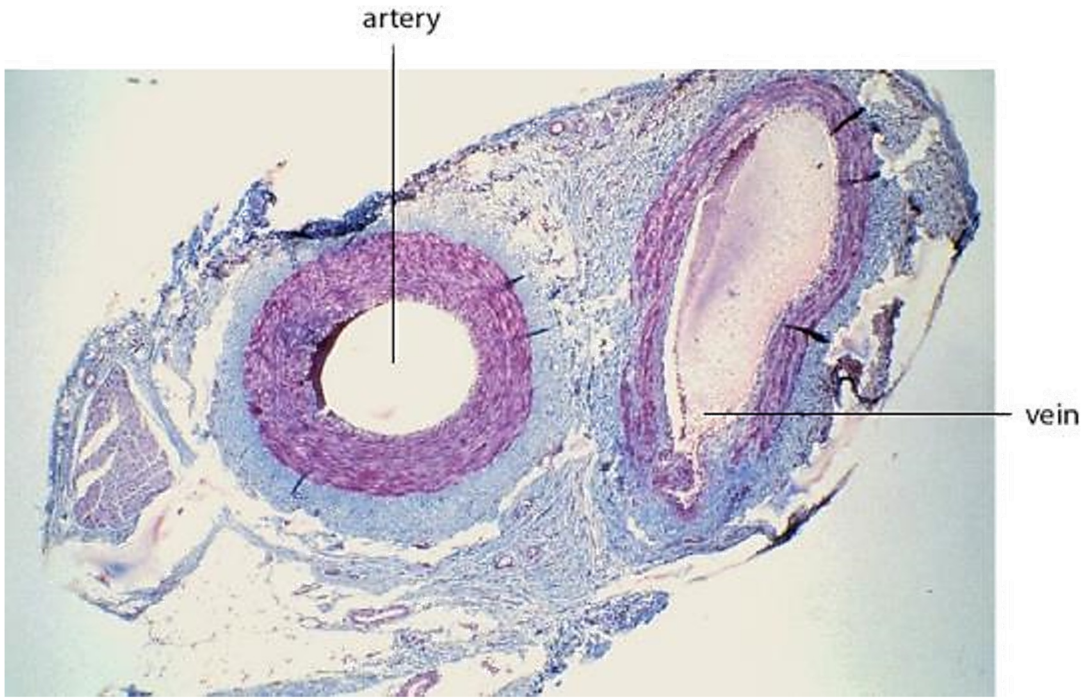
HELPFUL NOTES

Blood is never blue. It is coloured blue in this diagram to represent deoxygenated blood.

Arteries

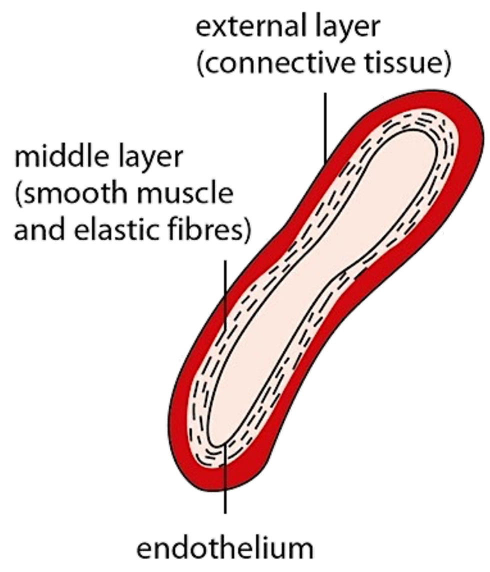


Cross-section of an artery

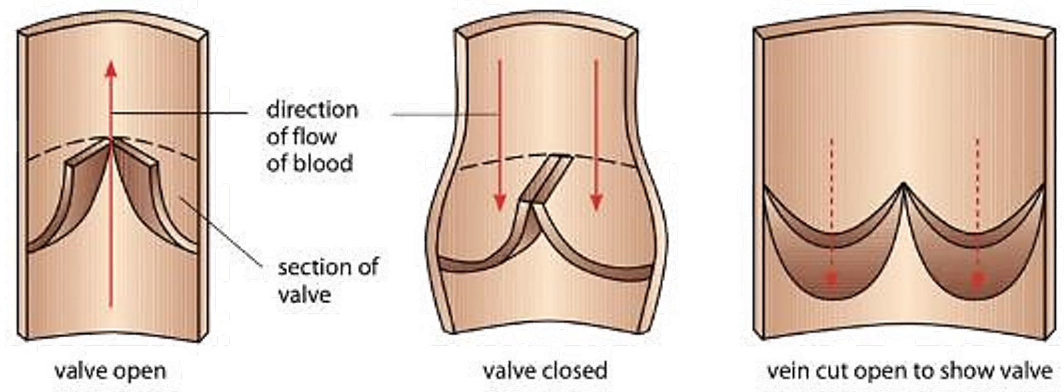


Light micrograph of transverse section (T.S) through an artery and a vein (20× magnification)

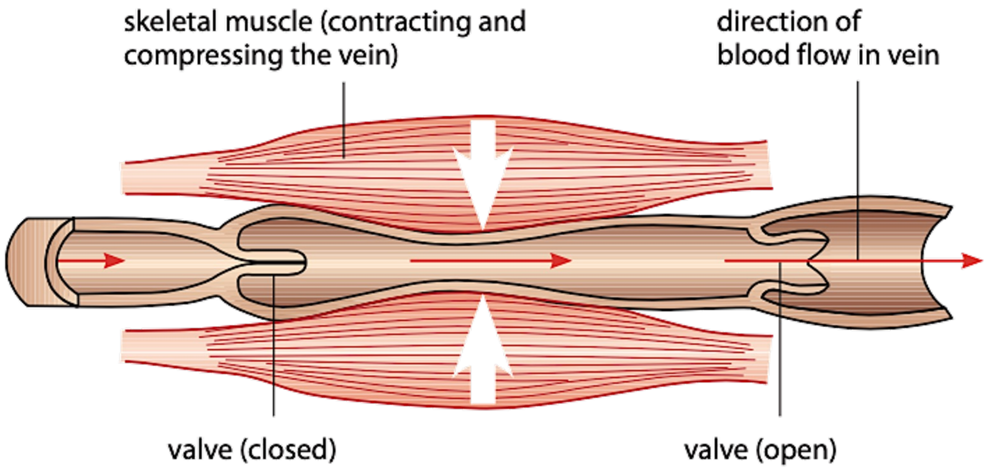
Veins



Cross-section of a vein

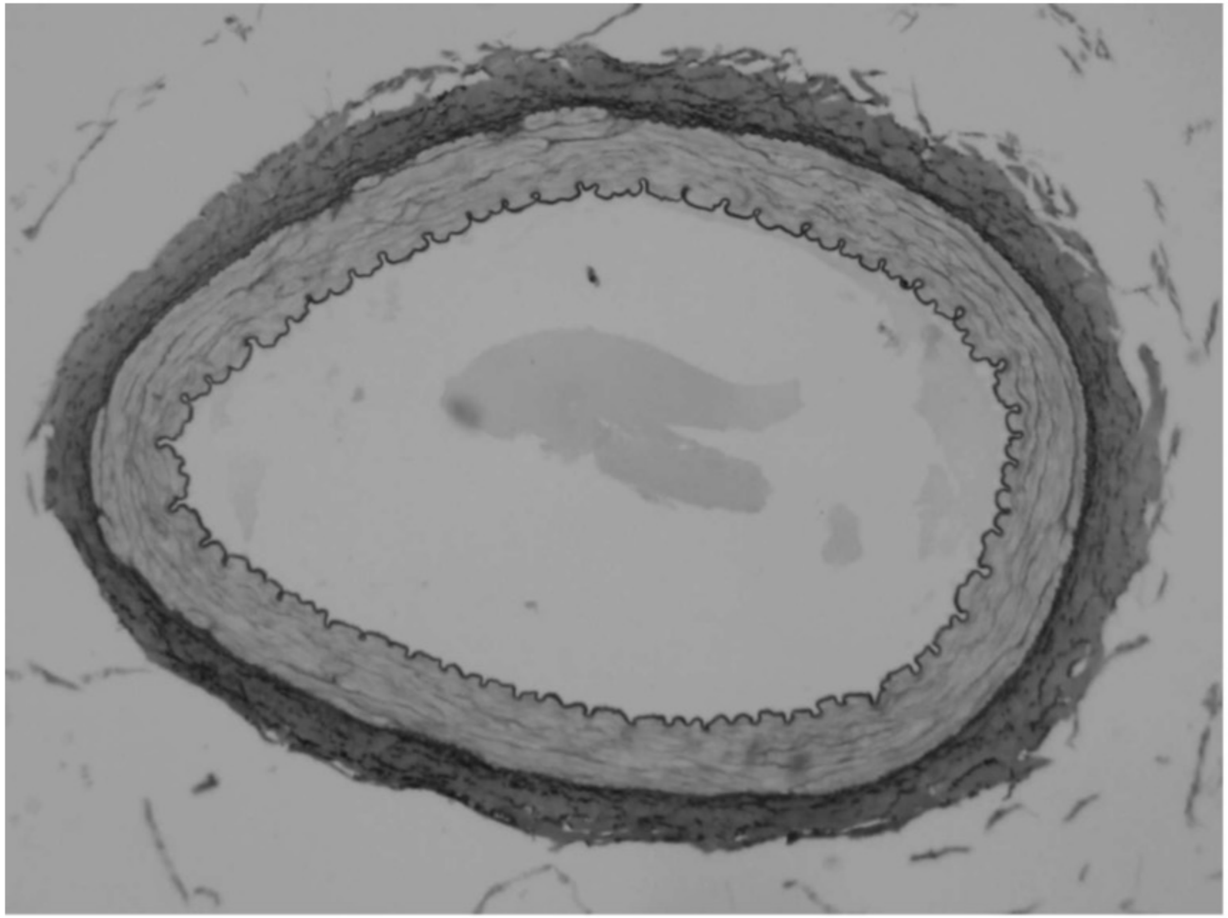


How the semilunar valve in a vein works



How skeletal muscles help blood flow in the vein

The photograph below shows a cross-section of an artery.



Explain how the structure of an artery, as shown in the photograph above, is related to its functions.

1 thick, wall;

2 withstands (blood) pressure;

3 muscular tissue;

4 vasoconstriction/ vasodilation;

5 elastic (tissue);

6 recoils to maintain (blood) pressure/ smoothes out blood flow;

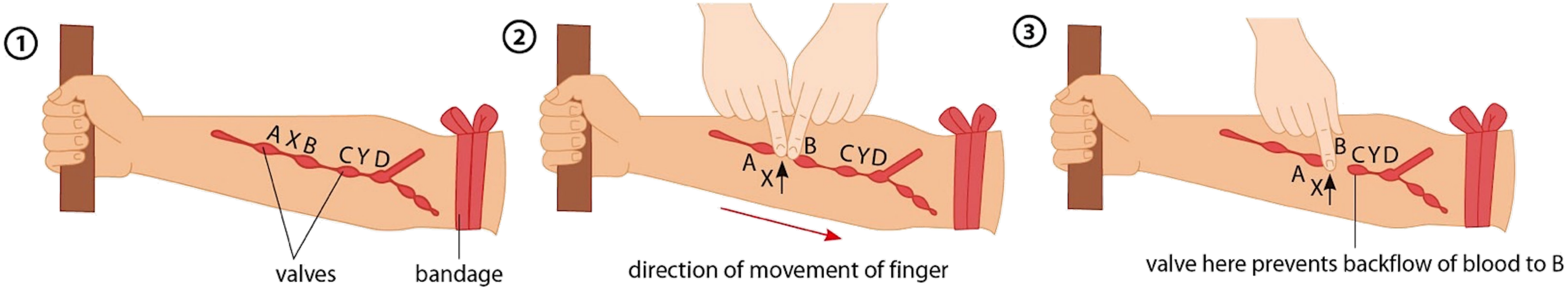
7 small lumen;

8 maintains (blood) pressure;

9 fibrous tissue;

10 maintains shape

Demonstrating the presence of valves in veins

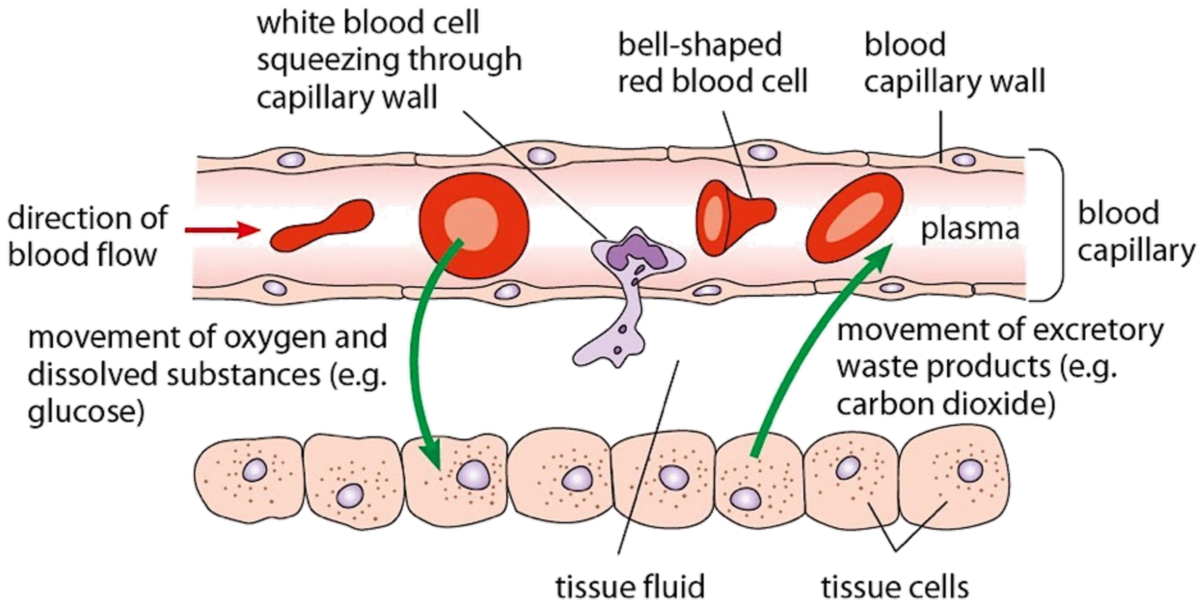


Action of valves in the veins of an arm

Tissue fluid

The tiny spaces between tissue cells contain a colourless liquid known as **tissue fluid**. Tissue cells are bathed with tissue fluid.

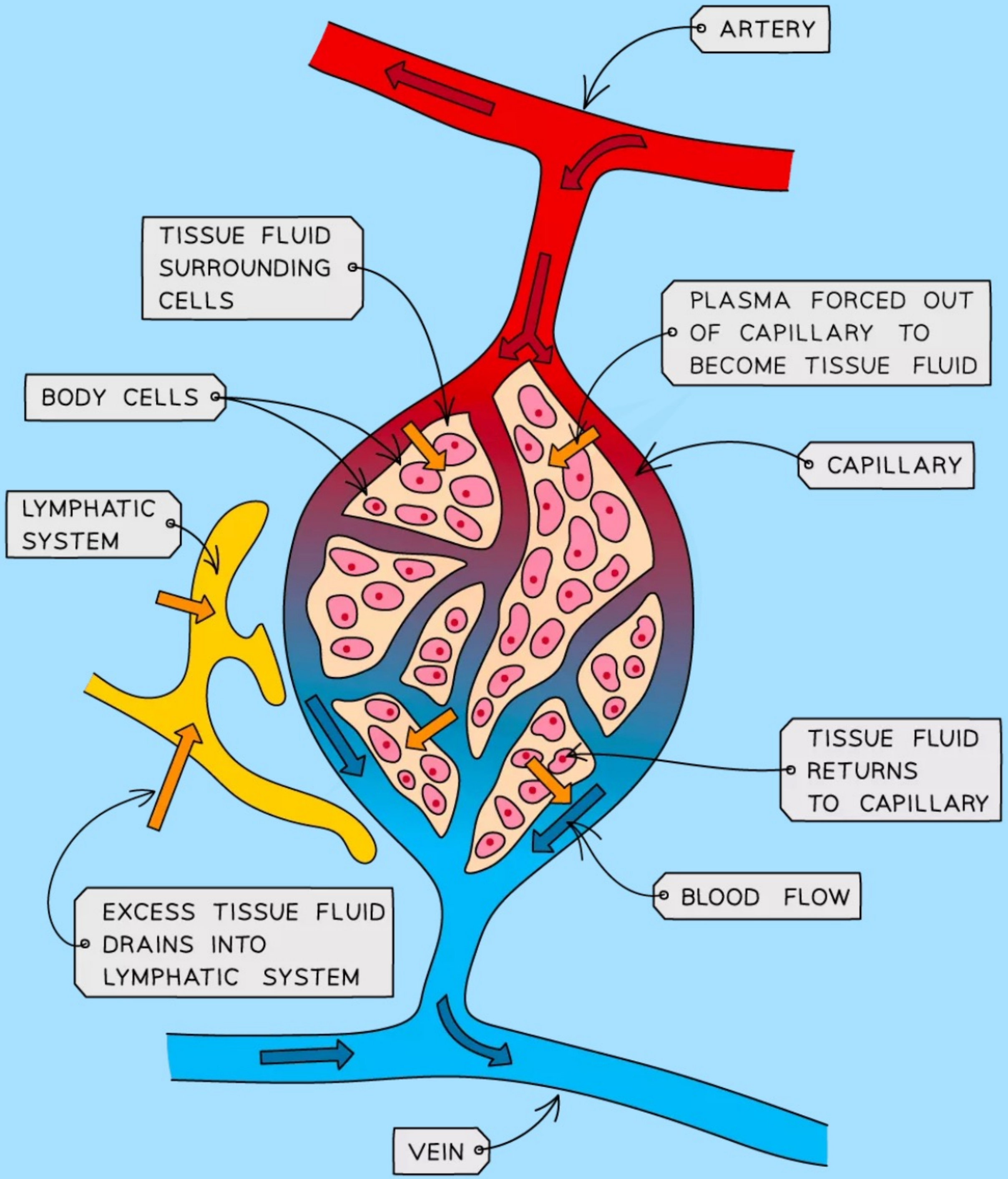
How are substances transferred between capillaries and tissue cells?



The relationship between a blood capillary, tissue fluid and tissue cells

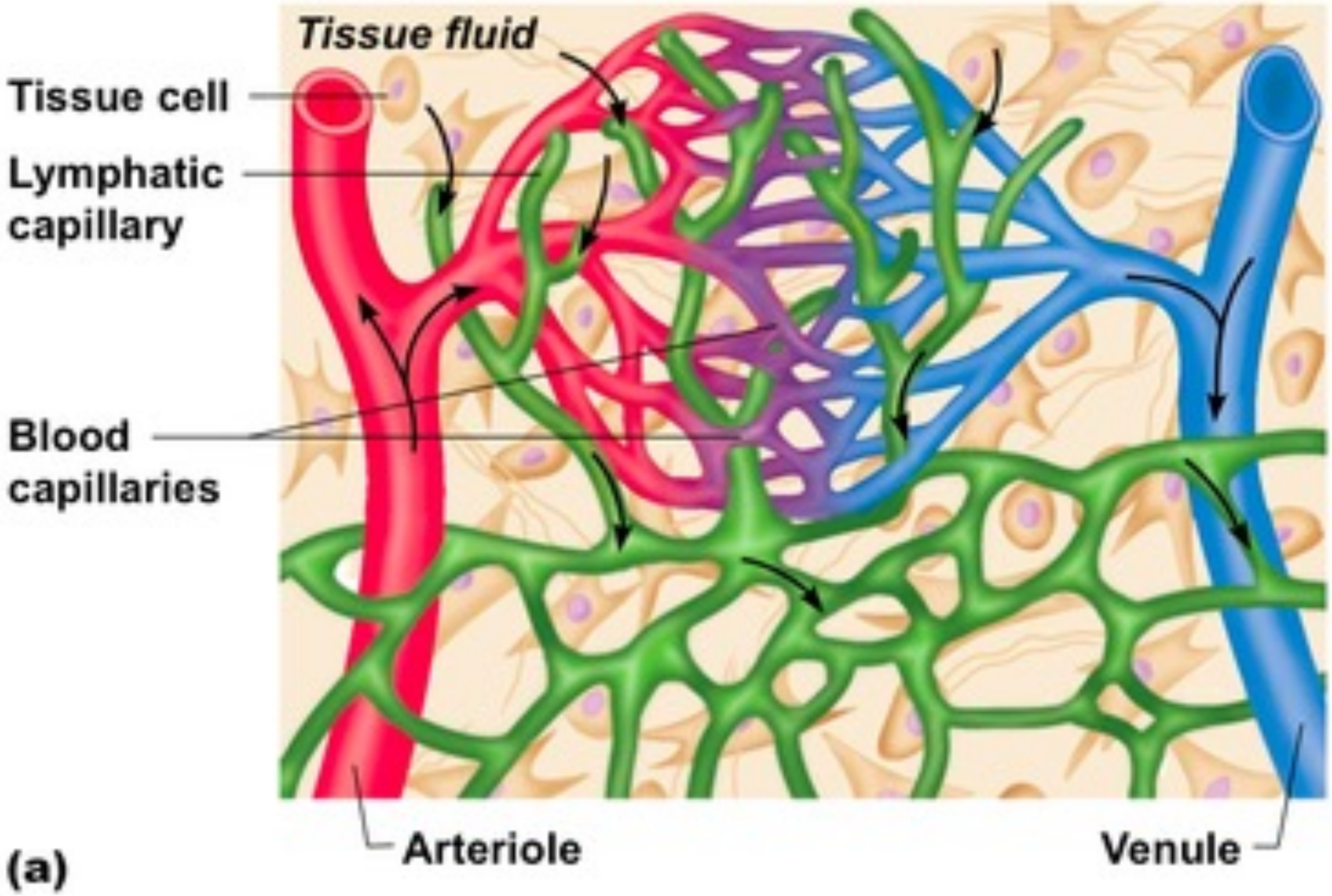
Lymph Fluid

- The walls of the capillaries are so thin that water, dissolved solutes and dissolved gases easily leak out of them / pass through the walls from the plasma into the **tissue fluid** surrounding the cells
- Cells **exchange materials** (such as water, oxygen, glucose, carbon dioxide, mineral ions) **across their cell membranes** with the tissue fluid surrounding them by diffusion, osmosis or active transport
- More fluid leaks out of the capillaries than is returned to them, and this excess of leaked fluid surrounding the capillaries then passes into the lymphatic system, becoming **lymph fluid**


ARTERY
TISSUE FLUID SURROUNDING CELLS
PLASMA FORCED OUT OF CAPILLARY TO BECOME TISSUE FLUID
BODY CELLS
CAPILLARY
LYMPHATIC SYSTEM
TISSUE FLUID RETURNS TO CAPILLARY
BLOOD FLOW
EXCESS TISSUE FLUID DRAINS INTO LYMPHATIC SYSTEM
VEIN

Lymph Vessels & Nodes

- The lymphatic system is formed from a **series of tubes which flow from tissues back to the heart**
- It connects with the blood system near to the heart, where lymph fluid is **returned** to the blood plasma
- **Lymph nodes** are small clusters of lymphatic tissue found throughout the lymphatic system, especially in the neck and armpits
- Large numbers of **lymphocytes** are found in lymph nodes
- Tissues associated with the lymphatic system, such as bone marrow, produce these **lymphocytes**
- Lymphocytes play an important role in **defending the body against infection**


Tissue fluid
Tissue cell
Lymphatic capillary
Blood capillaries
Arteriole
Venule
(a)

How are substances transferred between capillaries and tissue cells?

Comparison of arteries, veins and capillaries

	Arteries	Veins	Capillaries
Structure	Have thick muscular walls with much elastic tissue	Have thin muscular walls with little elastic tissue	Have one-cell thick walls with no muscular or elastic tissue
	Have small lumen relative to diameter	Have large lumen relative to diameter	Have large lumen relative to diameter
	Semilunar valves absent	Semilunar valves present	Semilunar valves absent
Function	Carry blood away from the heart	Carry blood towards the heart	Link arteries to veins
	Carry oxygenated blood (except for pulmonary arteries which carry deoxygenated blood from the heart to the lungs)	Carry deoxygenated blood (except for the pulmonary veins which carry oxygenated blood from the lungs to the heart)	Blood changes from oxygenated at the arteriole end to deoxygenated at the venule end

How are substances transferred between capillaries and tissue cells?

Comparison of arteries, veins and capillaries

	Arteries	Veins	Capillaries
Flow	Blood under high pressure	Blood under low pressure	Pressure of blood reduces as blood flows from arteriole to venule end
	Blood moves in pulses, reflecting the rhythmic, pumping action of the heart	No pulse	No pulse
	Blood flows rapidly	Blood flows slowly	Blood flows slowly

Let's Practise 9.1 and 9.2

1 State the main function(s) of the following.

(a) S A lymphocyte

(b) S A phagocyte

(c) A red blood cell

(d) A platelet

2 **(a)** Why do large animals such as mammals need a transport system?

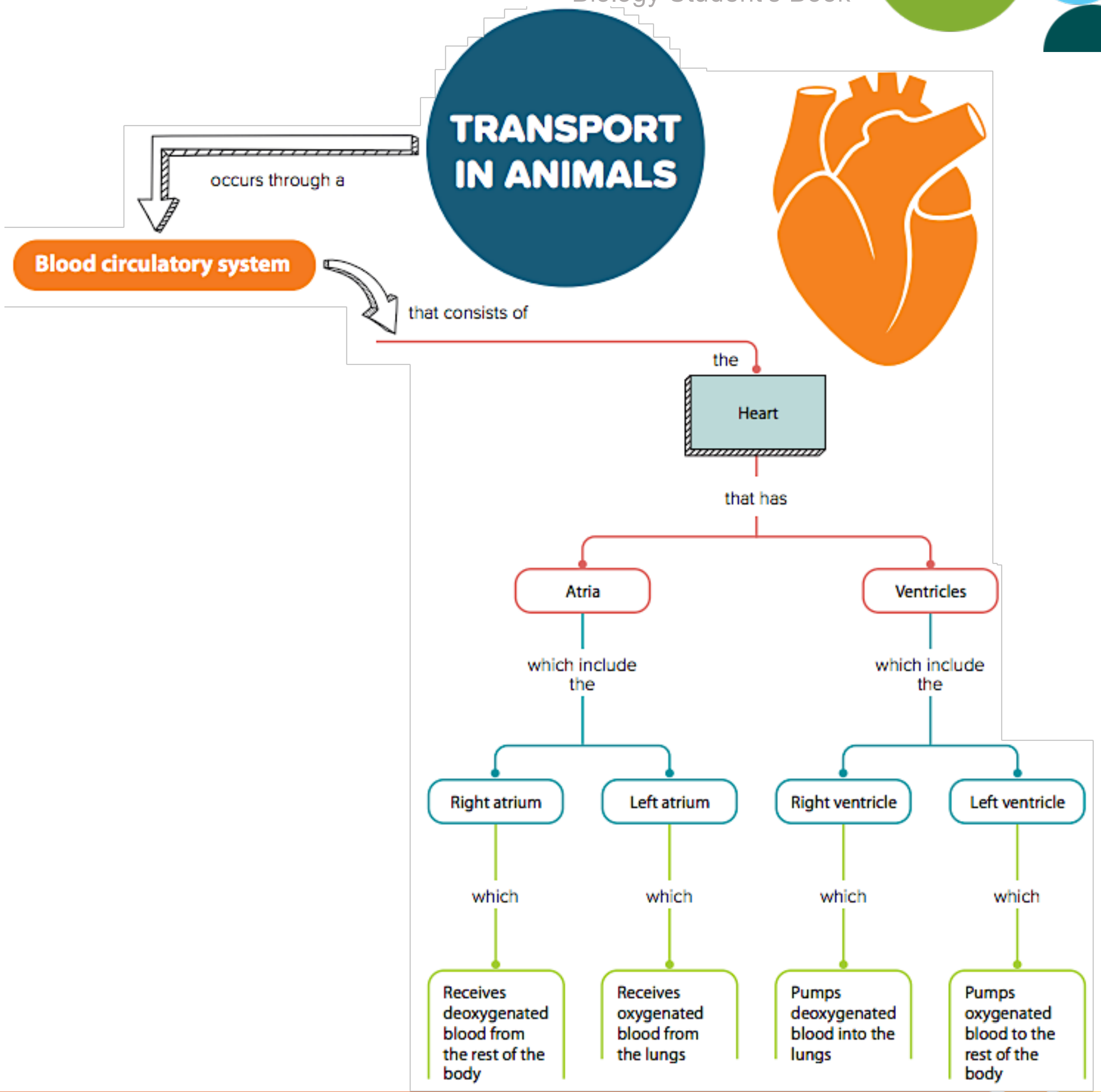
(b) Name the type of transport system found in mammals.

(c) Describe **three** structural differences between arteries, veins and capillaries.

9.4 The Heart

In this section, you will learn the following:

- Identify the structures of the mammalian heart.
- S Explain and compare the relative thickness of the muscle walls of the left and right ventricles and the atria.
- S Explain the importance of the septum.
- S Identify the atrioventricular and semilunar valves in the mammalian heart.
- S Describe how the heart functions in terms of contraction of the atria and ventricles, and action of the valves.
- State that ECG, pulse rate and listening to the sound of valves closing may be used to monitor the activity of the heart.
- Identify the main blood vessels to and from the heart, lungs and kidney.
- S Identify the hepatic artery, hepatic veins and hepatic portal vein as the main blood vessels to and from the liver.
- Investigate and describe the effect of physical activity on heart rate.
- S Explain the effect of physical activity on the heart rate.



9.2 Heart

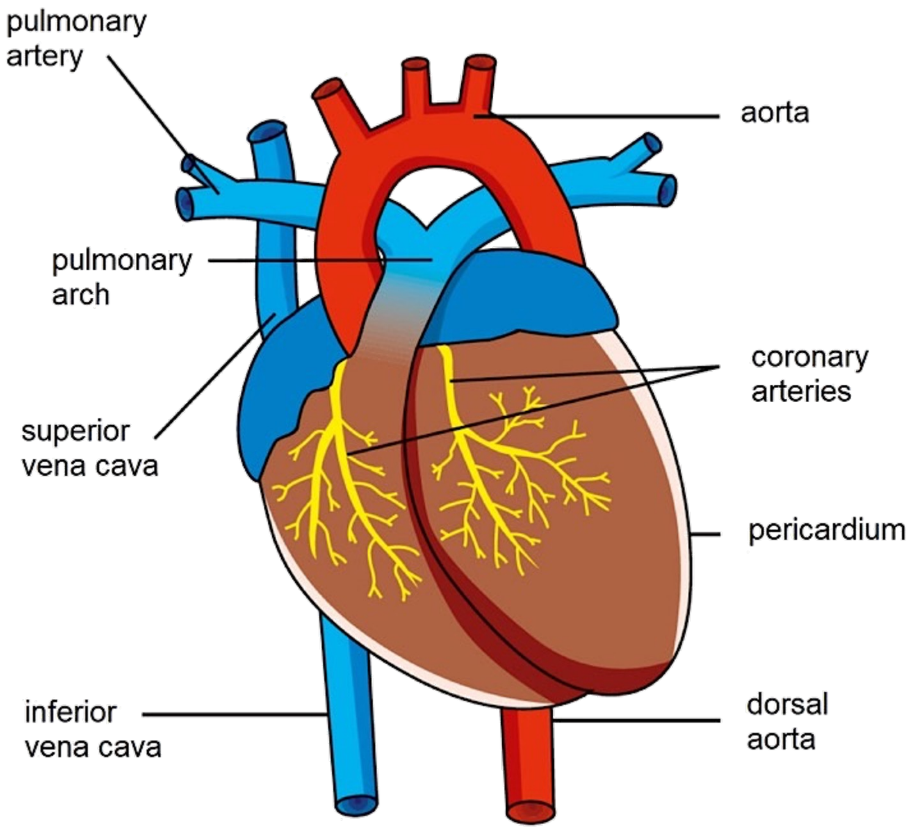
Core

1. Identify in diagrams and images the structures of the mammalian heart, limited to: muscular wall, septum, left and right ventricles, left and right atria, one-way valves and coronary arteries

Supplement

7. Identify in diagrams and images the atrioventricular and semilunar valves in the mammalian heart
8. Explain the relative thickness of:
   (a) the muscle walls of the left and right ventricles
   (b) the muscle walls of the atria compared to those of the ventricles
9. Explain the importance of the septum in separating oxygenated and deoxygenated blood
10. Describe the functioning of the heart in terms of the contraction of muscles of the atria and ventricles and the action of the valves

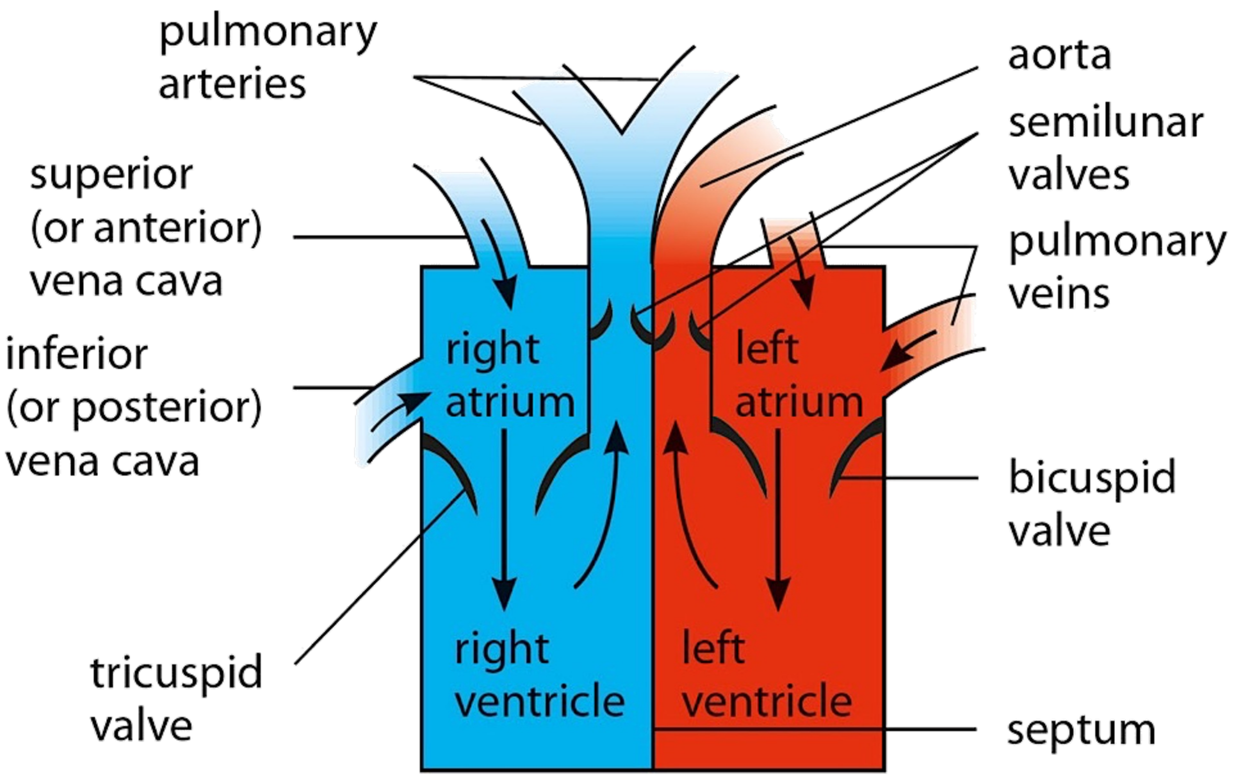
Structure of the heart



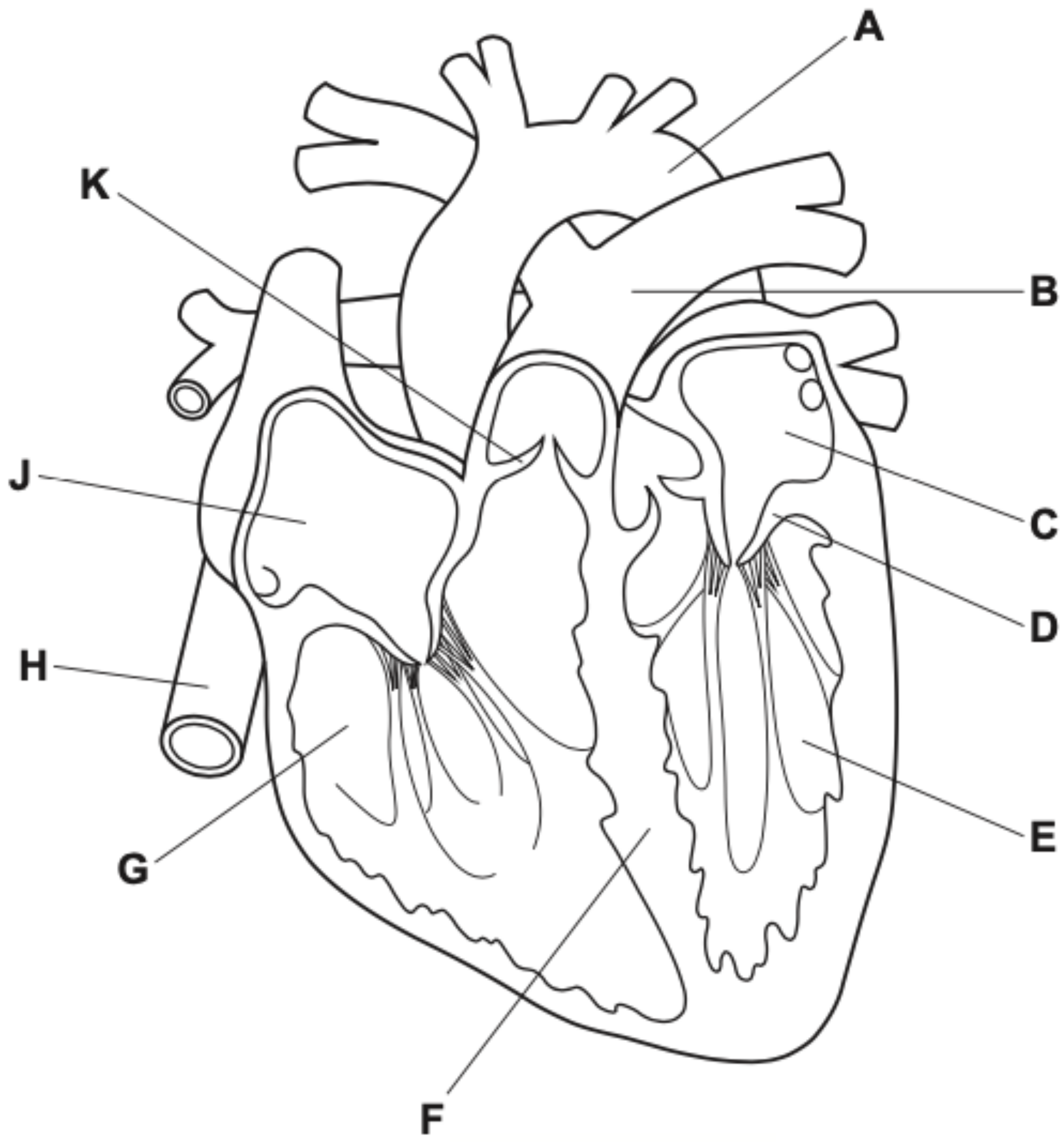
The mammalian heart and its associated blood vessels

S

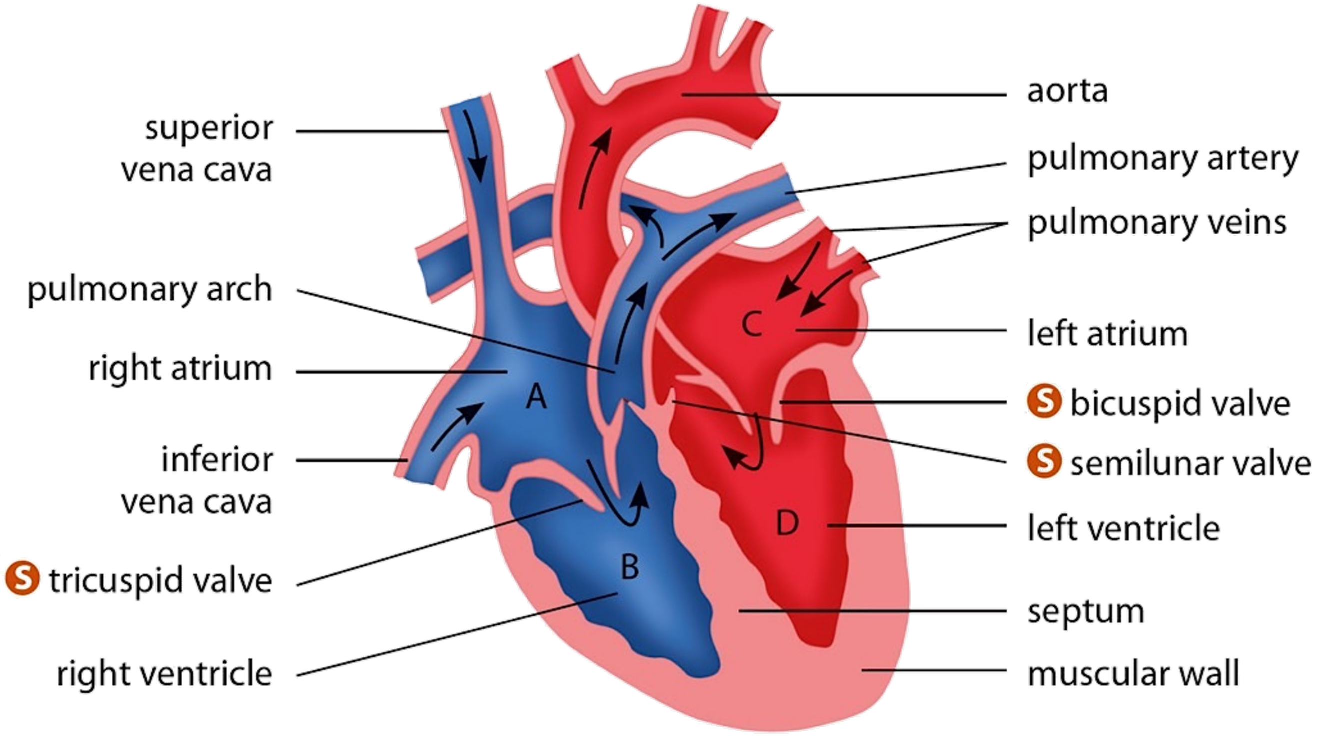
The mammalian heart



Simplified representation of a mammalian heart


A
B
C
D
E
F
G
H
J
K

What path does blood take through the heart?

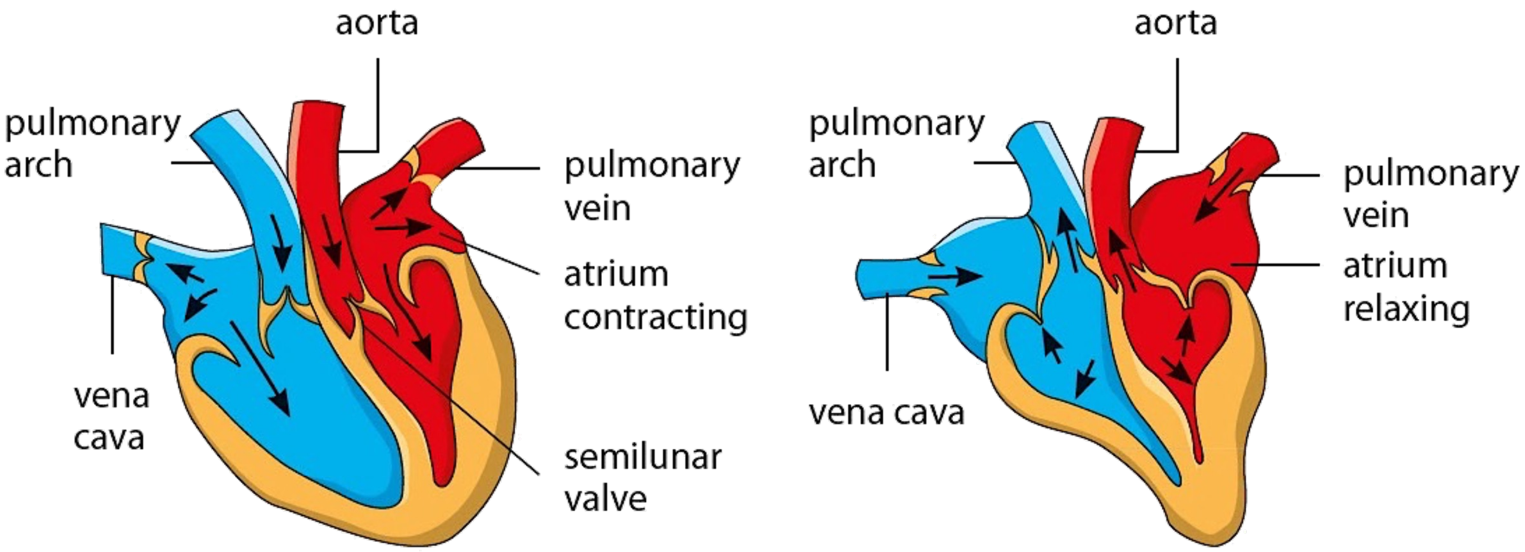


Path of blood through the heart

HELPFUL NOTES

When you examine a heart, the right side of the heart will be facing your left and vice versa.

How the heart works



Ventricular relaxation

Ventricular contraction

Mode of action of the heart

HELPFUL NOTES

A systole and a diastole make up one heartbeat.

Similarly, one "lub" and one "dub" sound make up one heartbeat. One heartbeat lasts about 0.8 s.

2 State that blood is pumped away from the heart in arteries and returns to the heart in veins

3 State that the activity of the heart may be monitored by: ECG, pulse rate and listening to sounds of valves closing

4 Investigate and describe the effect of physical activity on the heart rate

5 Describe coronary heart disease in terms of the blockage of coronary arteries and state the possible risk factors including: diet, lack of exercise, stress, smoking, genetic predisposition, age and sex

6 Discuss the roles of diet and exercise in reducing the risk of coronary heart disease

11 Explain the effect of physical activity on the heart rate

Blood Pressure

Blood pressure is the force of the blood exerted on the walls of the blood vessels.

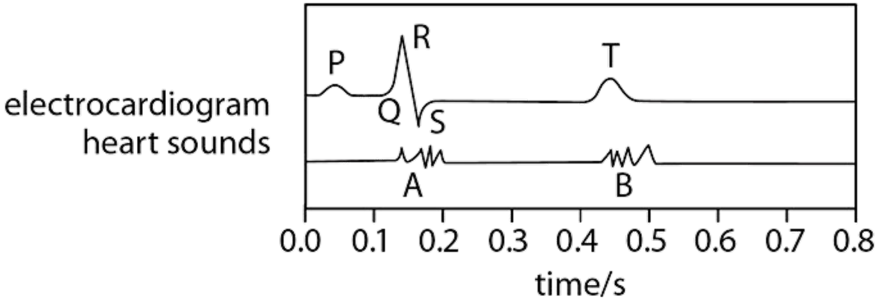
Using a sphygmomanometer to measure blood pressure

The pulse

A pulse is produced after every ventricular contraction.

Electrocardiogram (ECG)

An ECG monitors the functioning of the heart by measuring its electrical activity.



Normal heart activity in an ECG

Let's Investigate 9B

- The effect of gravity is exerted more strongly when you stand up. Hence, more effort is required to pump blood throughout the body and the pulse rate changes.
- During physical exercise, more energy is needed for muscle contraction. This leads to an increase in aerobic respiration.
- The heart will have to beat faster so that oxygen and glucose can be transported more quickly to the muscles to meet the increase in demand. This also allows for the carbon dioxide produced to be transported quickly from the muscles to the lungs for removal.
- The higher the fitness level, the lower the pulse rate.

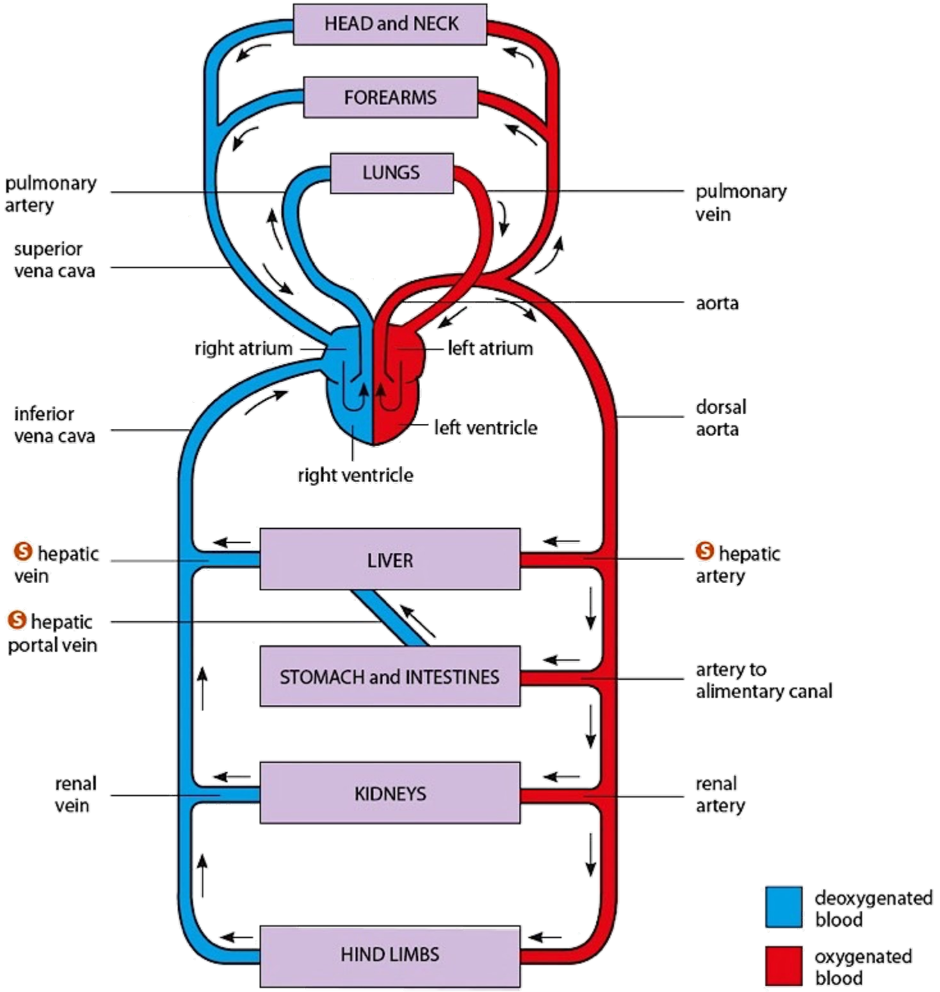
Physical activity increases the pulse rate so that oxygen and glucose can be transported more quickly to the muscles to meet the increase in demand.

Measuring pulse rate with two right fingers

LINK

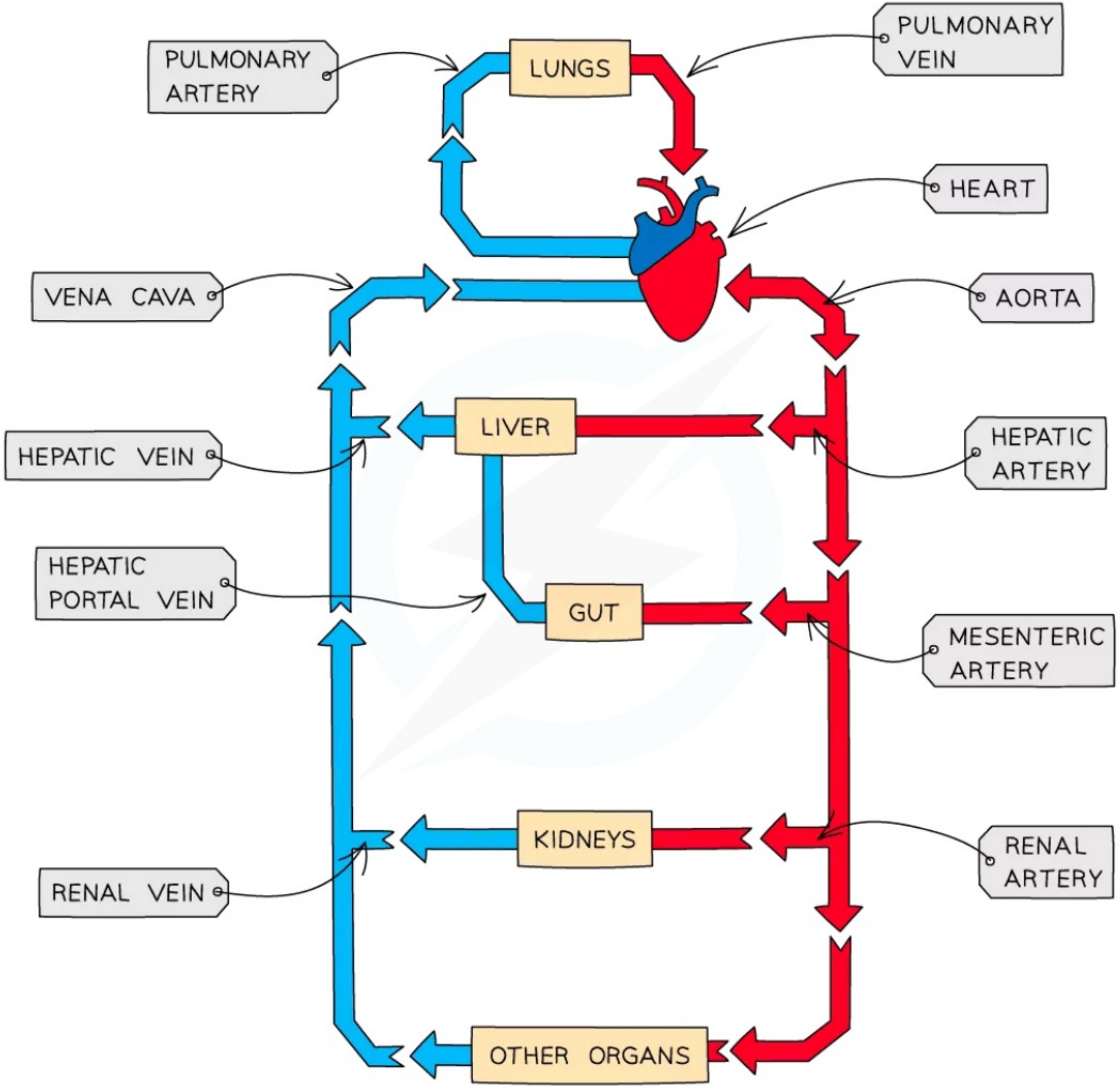
Find out about respiration in Chapter 12.

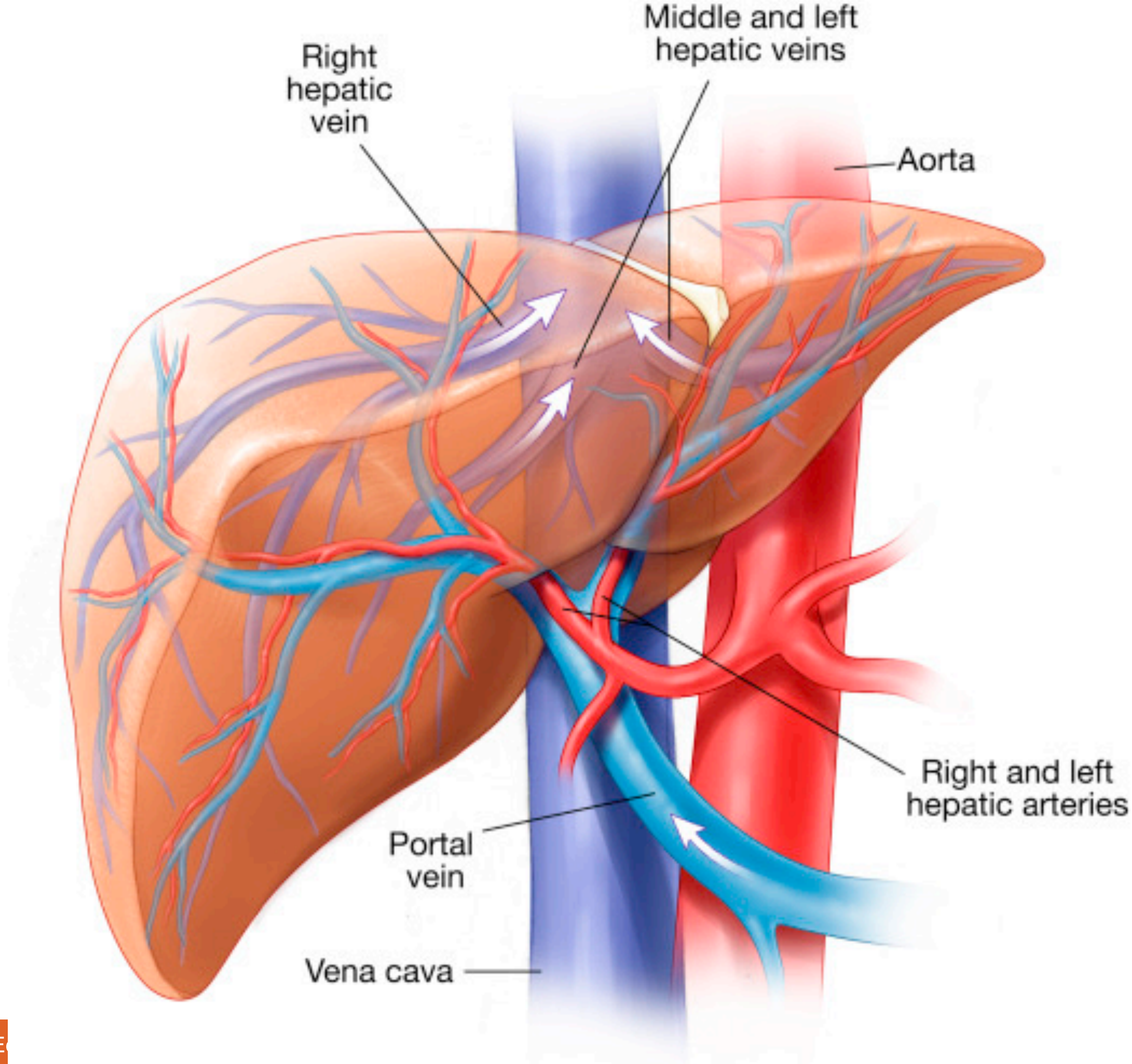
Arteries and veins of the human circulatory system

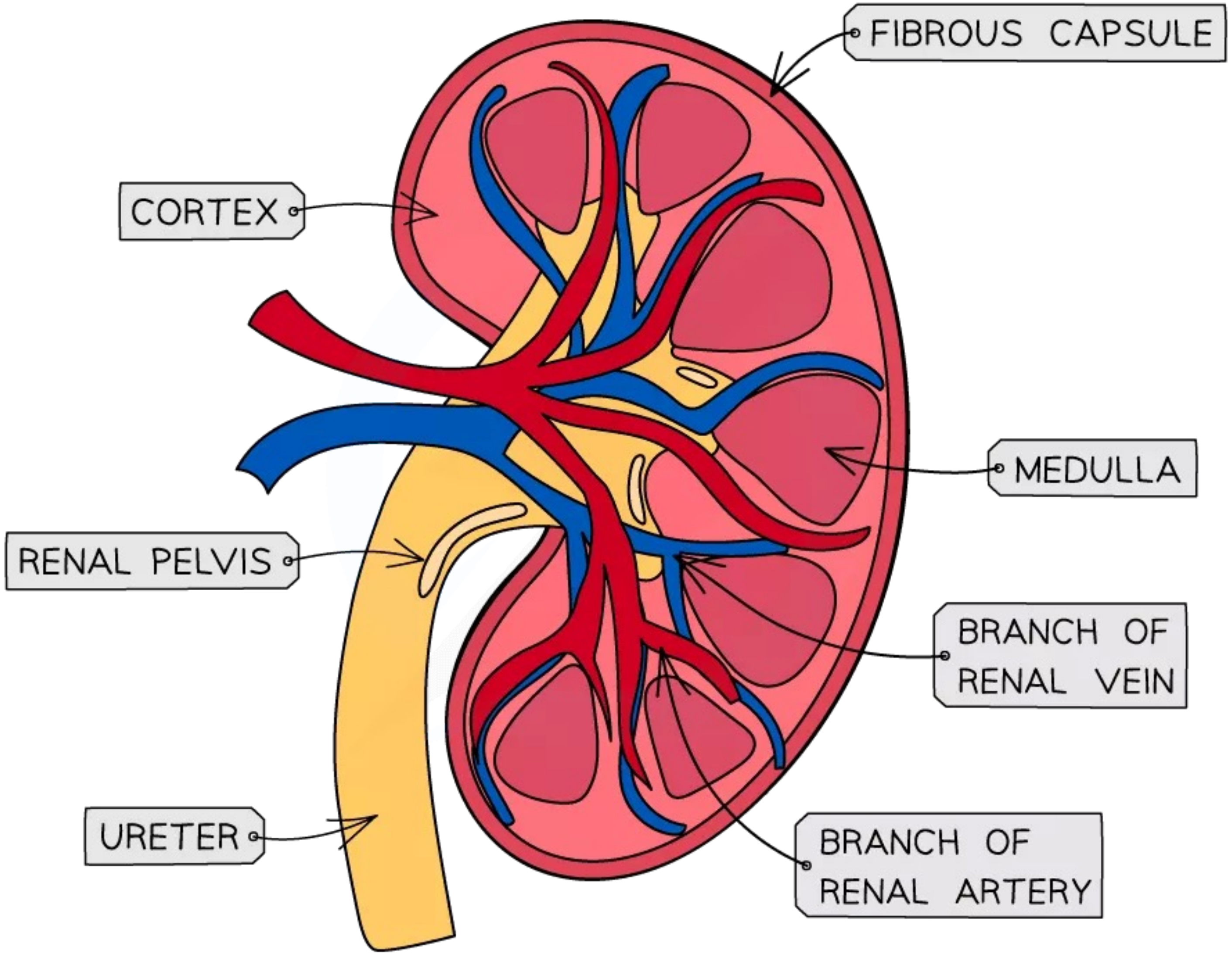


The main arteries and veins of the human circulatory system

Marshal Education



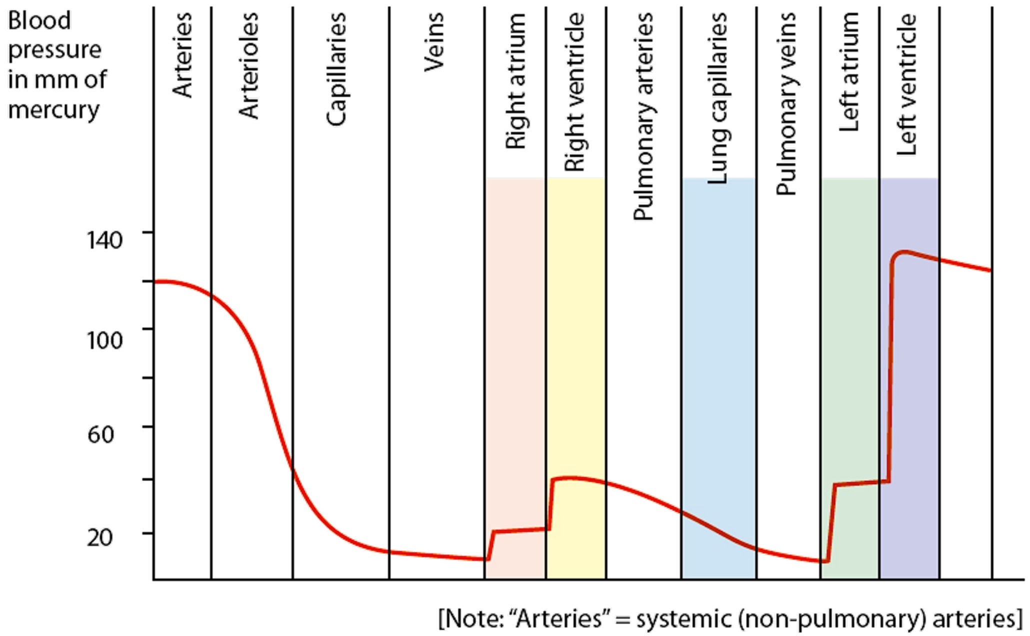

Right hepatic vein
Middle and left hepatic veins
Aorta
Right and left hepatic arteries
Portal vein
Vena cava


FIBROUS CAPSULE
CORTEX
MEDULLA
RENAL PELVIS
BRANCH OF RENAL VEIN
URETER
BRANCH OF RENAL ARTERY

Let's Practise 9.3 and 9.4

4 Use the information given in the figure to answer the following questions.

(a) In which region of the circulatory system is the pressure lowest?
(b) In which region of the circulatory system is there the greatest fall in pressure?
(c) How many times is the rise in pressure in the left ventricle greater than the rise in pressure in the left atrium?
(d) How is the blood pressure of a human measured?

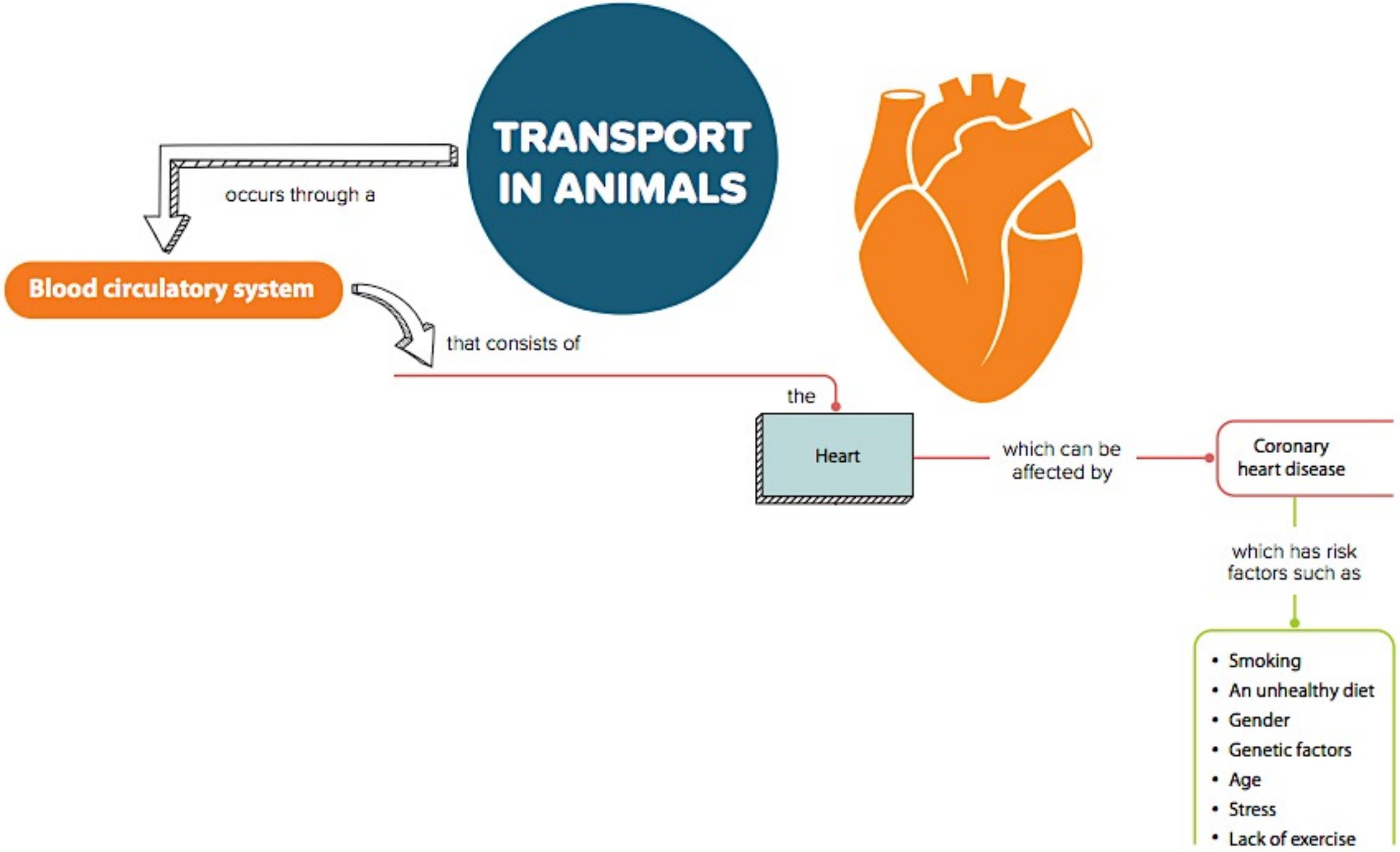


Diagrammatic representation of blood pressure in the circulatory system of a mammal

9.5 Coronary Heart Disease

In this section, you will learn the following:

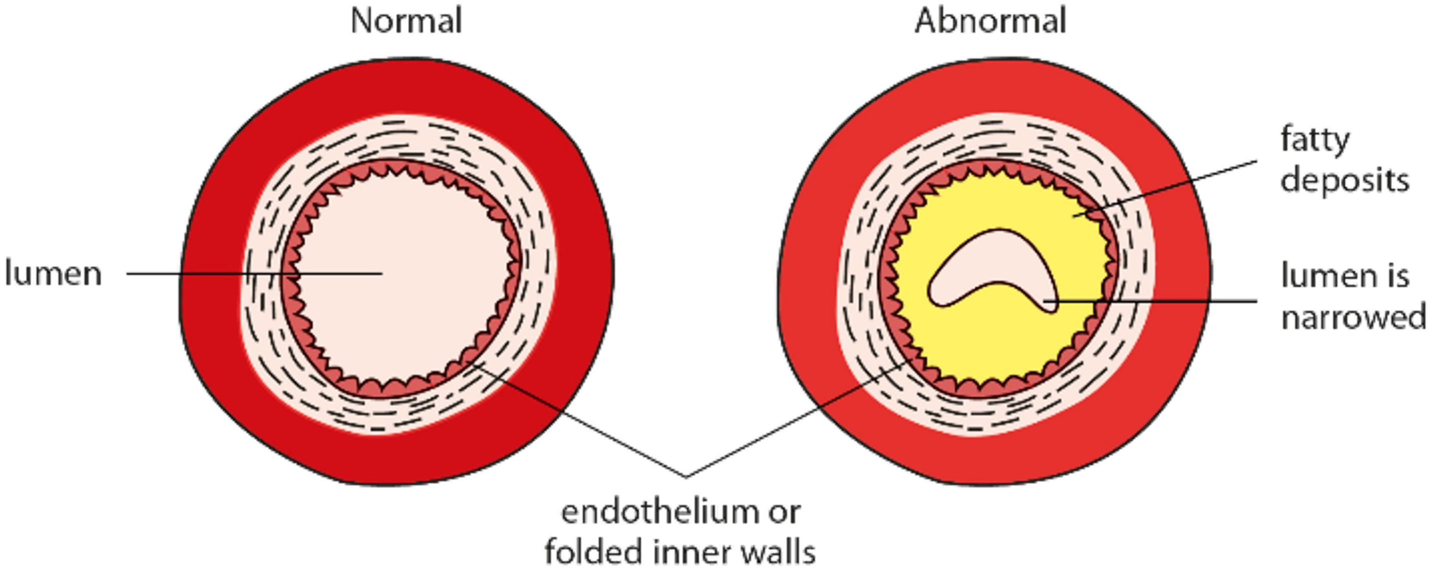
- Describe coronary heart disease and state some of the risk factors.
- Discuss how diet and exercise can reduce the risk of coronary heart disease.



What is a heart attack?

Blockage of the coronary arteries can cause a **heart attack**.

That region of heart muscles dies when it does not receive enough oxygen and nutrients. This becomes fatal when a large region of the heart is damaged.



Transverse sections of two arteries

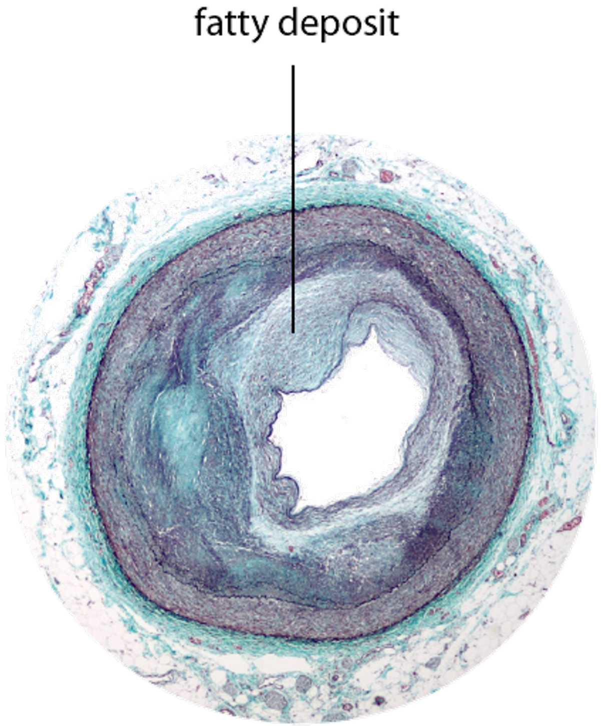
Causes of coronary heart disease

Atherosclerosis is the condition in which fatty substances are deposited on the inner surface of the coronary artery.

Thrombus is a blood clot that forms in an artery.

Factors that increase risk of coronary heart disease include

- a diet rich in cholesterol and saturated animal fats,
- emotional stress,
- smoking.



An artery with fatty substances deposited in lumen

Risk factors for heart disease

- Smoking
- Unhealthy diet
- Genetic factors
- Gender
- Age
- Stress
- Lack of exercise



Preventive measures against coronary heart disease

Proper diet

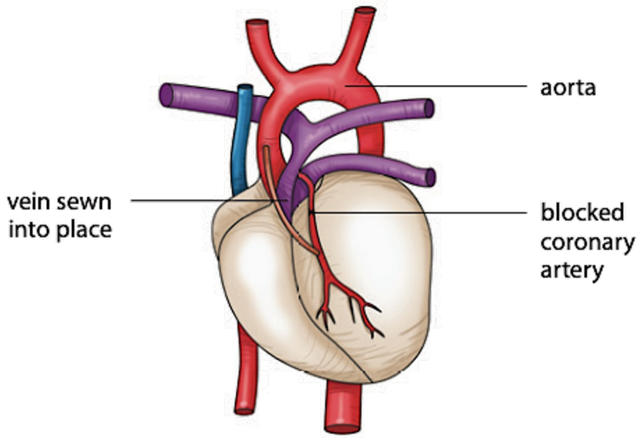
Proper stress management

Avoiding smoking

Regular physical exercise

Let's Practise 9.5

2 When a coronary artery is blocked, it can be replaced during surgery. A vein taken from another part of the patient's body is used to join the aorta to the coronary artery to bypass the blockage (Figure 9.27 on Student's Book p. 168). This surgery is known as coronary by-pass. Great care must be taken when the vein is sewn into place.

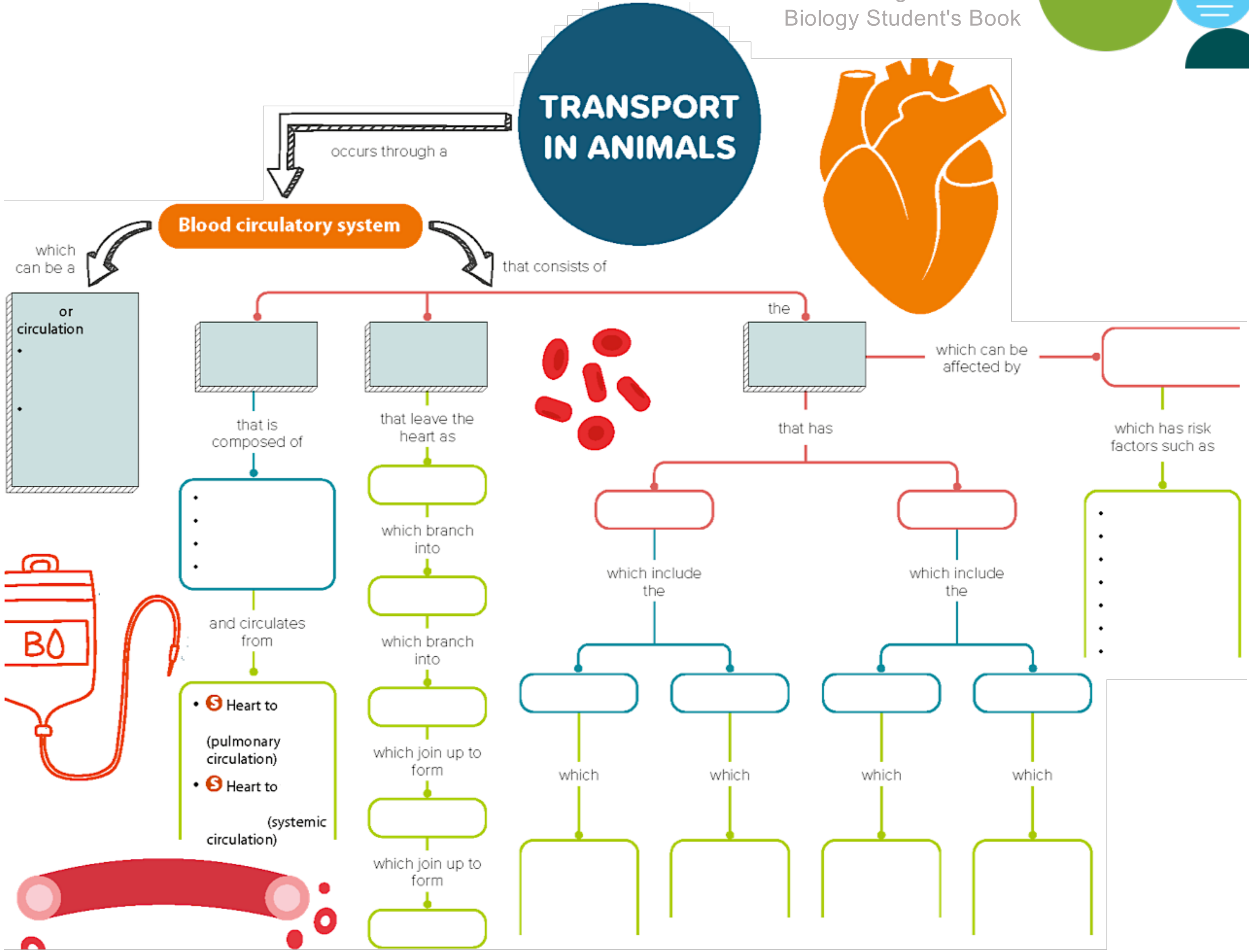


Coronary by-pass

Why must great care be taken to make sure that the vein is the correct way round?

What have you learned?

Can you draw your own mind map?



What have you learned?

Can you draw your own mind map?